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Mahler

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(54) **TAMPER DEVICE FOR A SCREED OF A WORKING MACHINE AND A METHOD FOR ADJUSTING A STROKE OF A TAMPER DEVICE FOR A SCREED OF A WORKING MACHINE**

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E01C 19/4866; B06B 1/167
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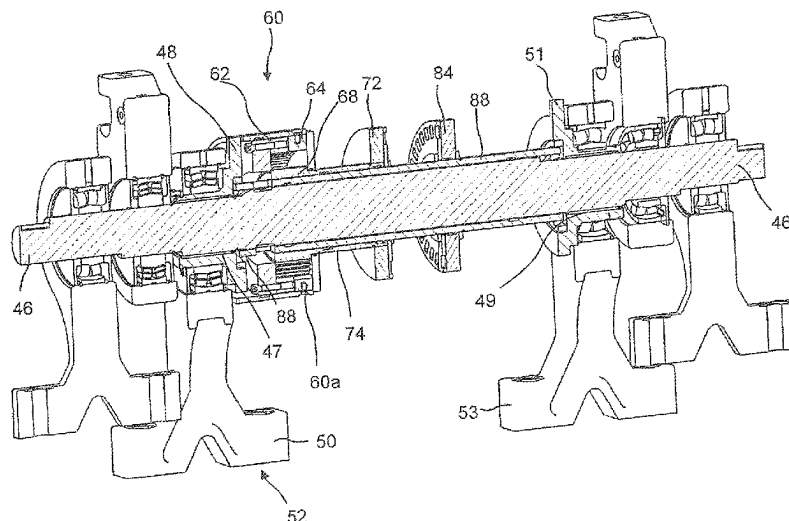
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(57) **ABSTRACT**

The invention relates to a tamper device for a screed of a working machine, in particular a paver. The device comprises a rotatable driveable tamper shaft with an eccentric section, an 5 eccentric bushing mounted on the eccentric section, and a connecting rod rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section. The device further comprises an inner-toothed hollow 10 wheel gear train for providing the rotational adjustment. The gear train is connected to the tamper shaft, to the eccentric bushing, and to a drive force receiving element configured to be able to receive a drive force for driving the innertoothed hollow wheel gear train when the tamper shaft is rotating.

27 Claims, 16 Drawing Sheets



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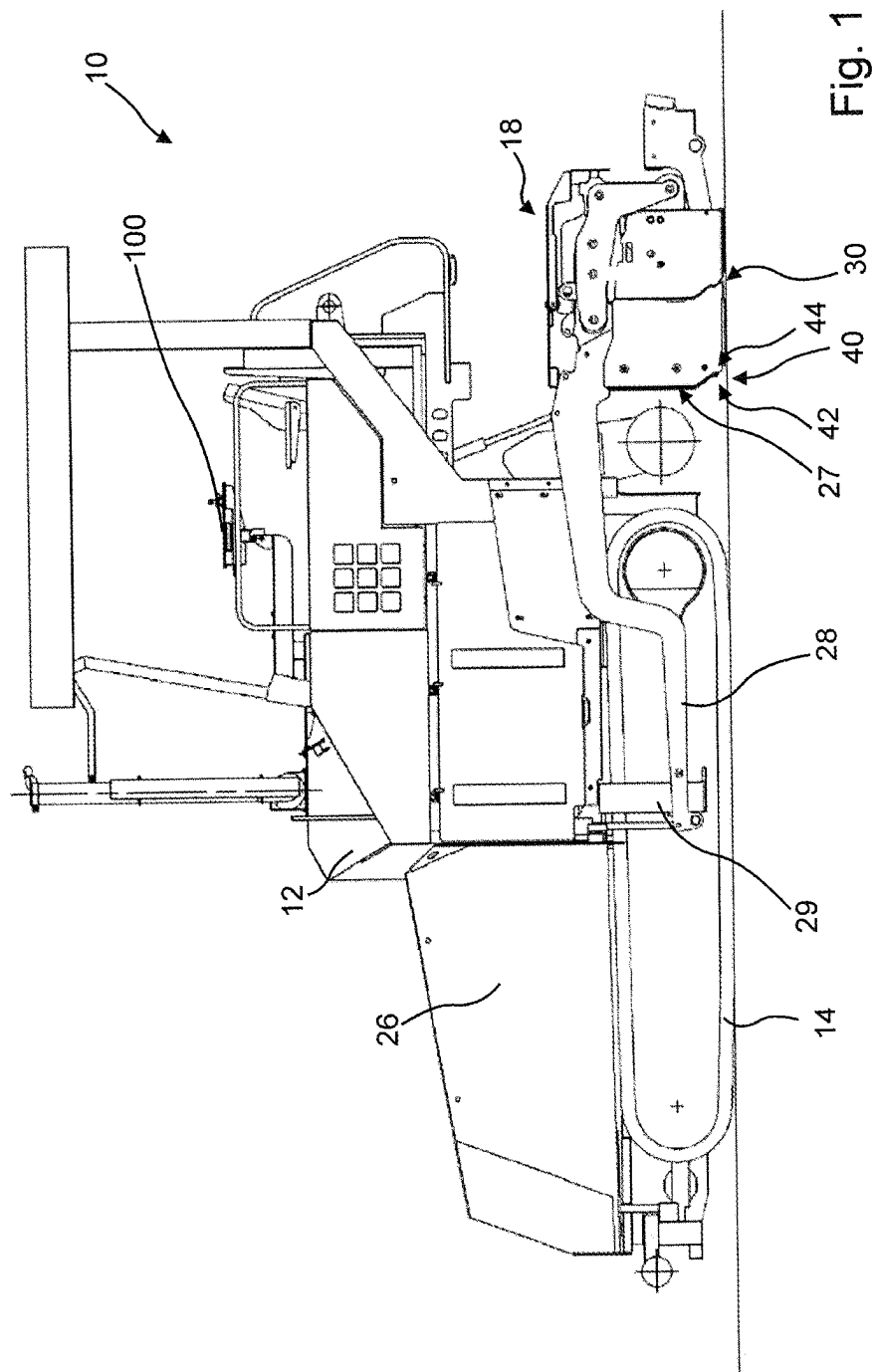
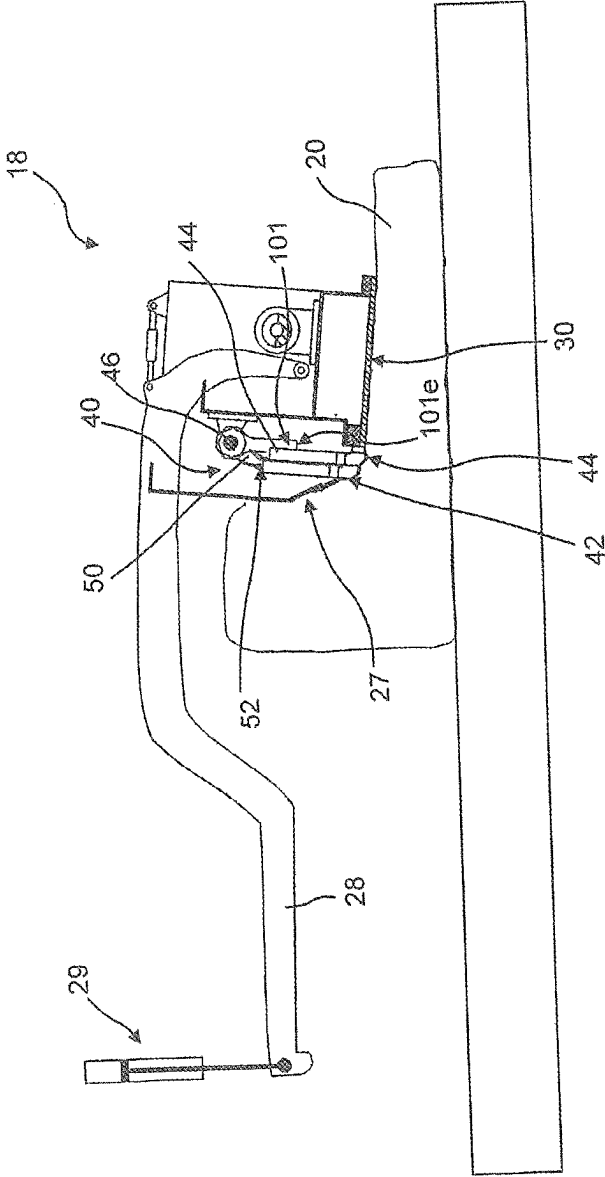


Fig. 2



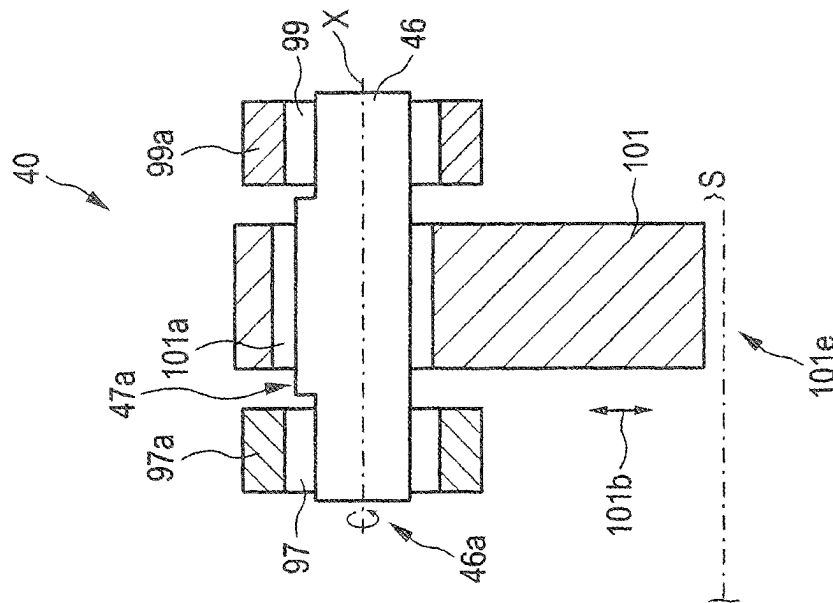


FIG. 2a

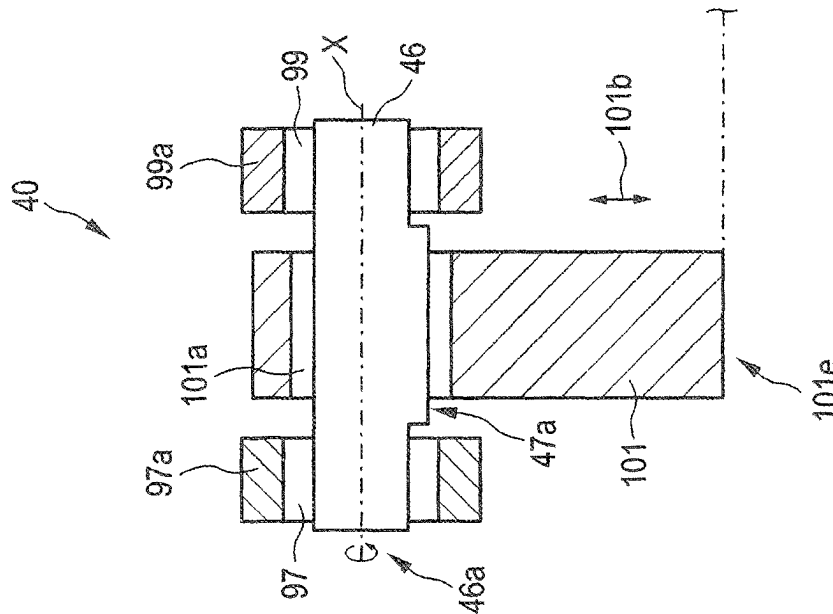


FIG. 2b

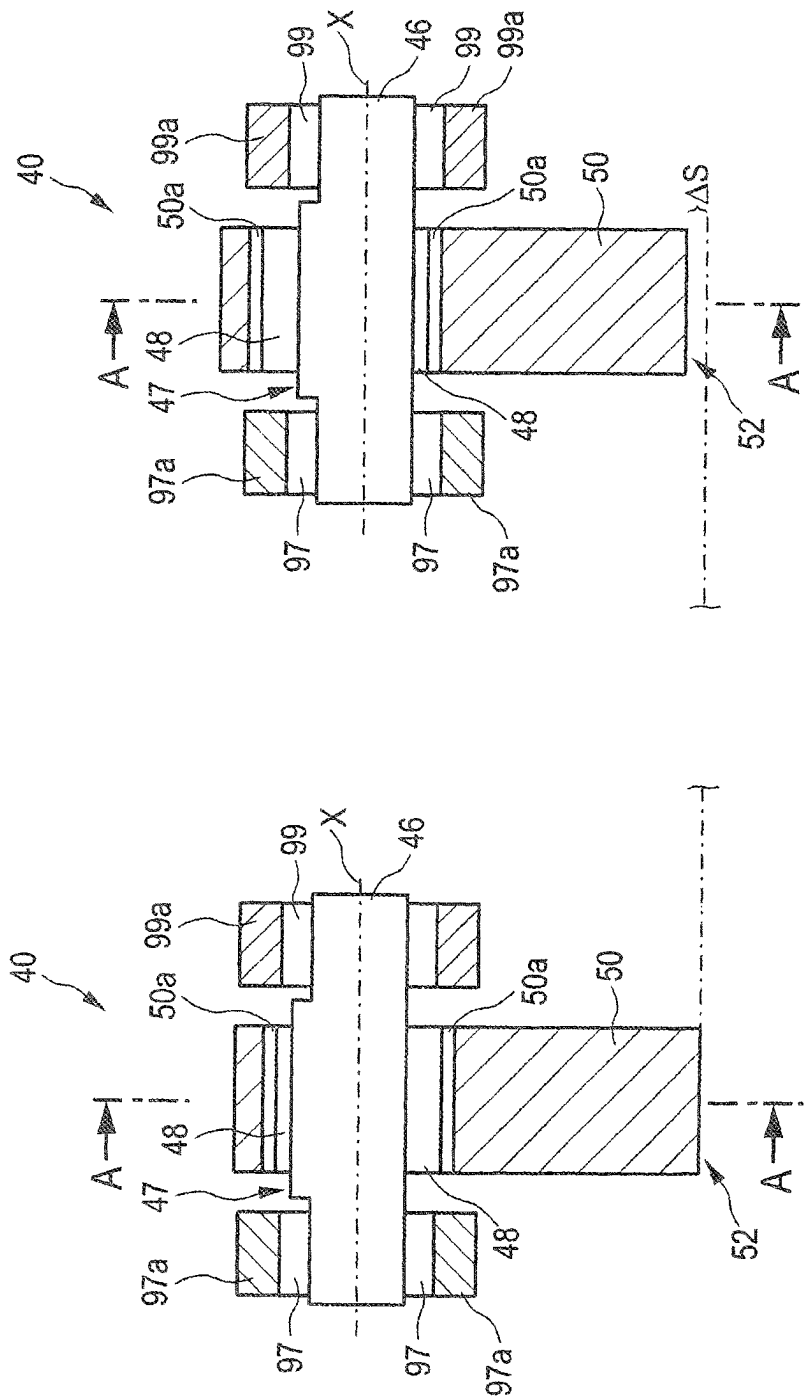
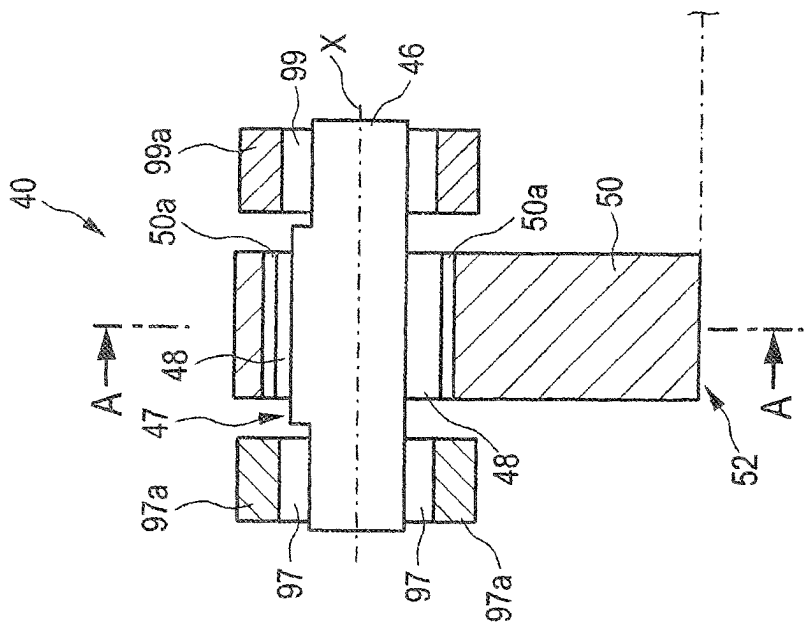


FIG. 2d



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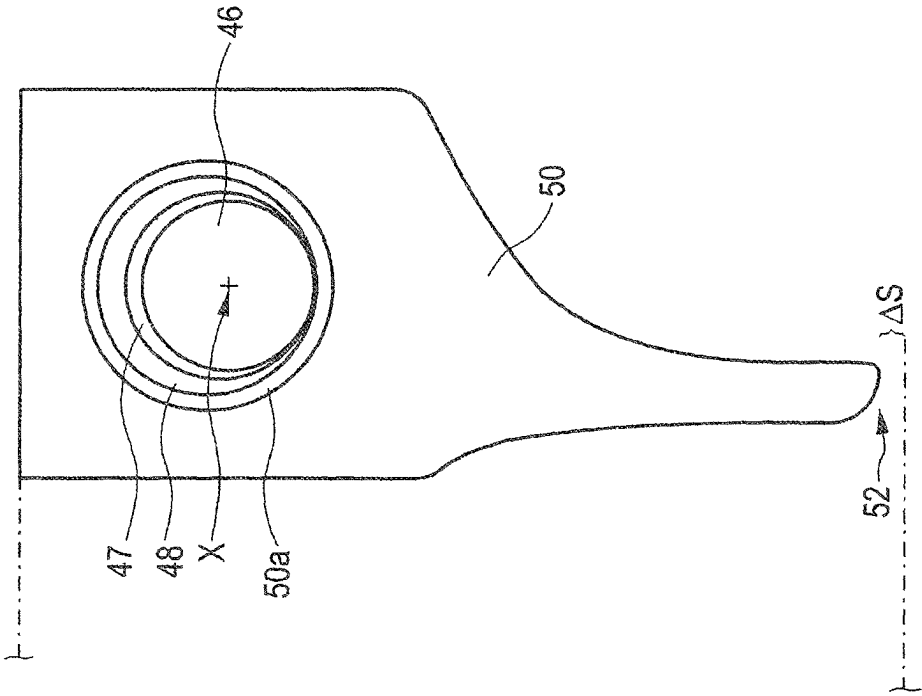


FIG. 2f

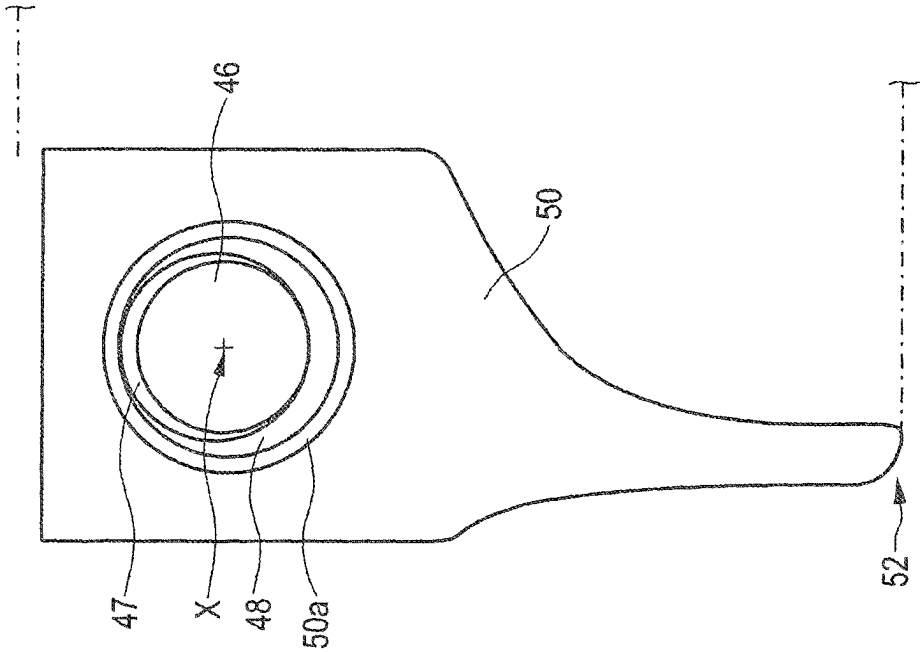


FIG. 2e

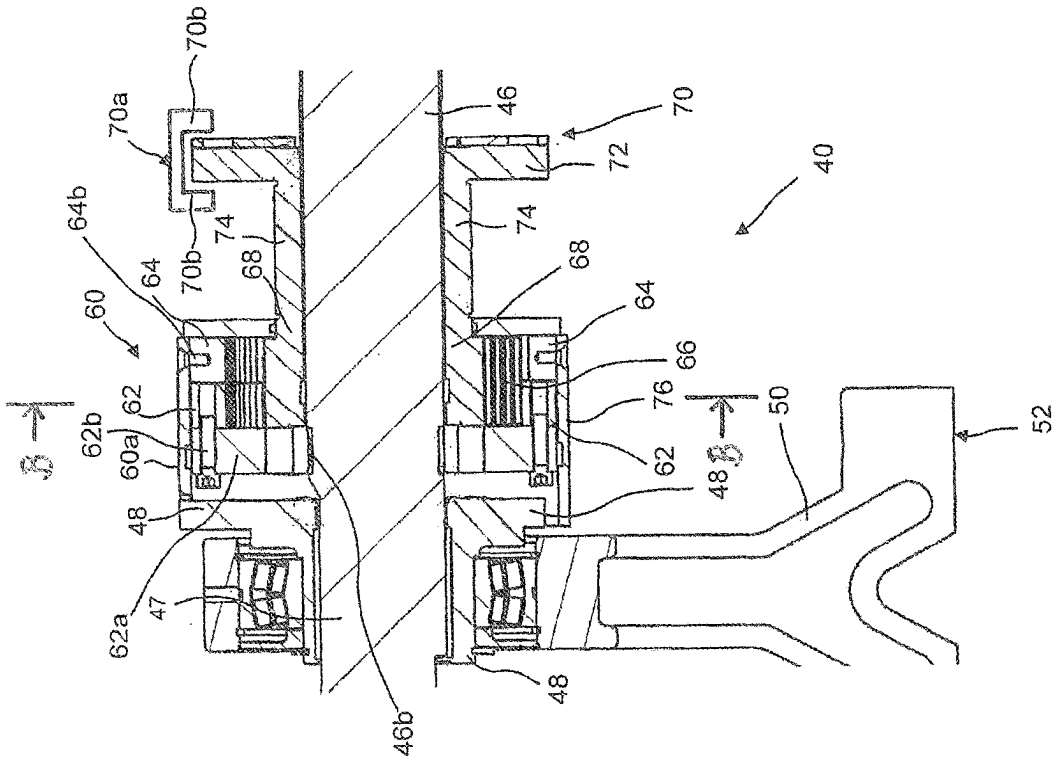
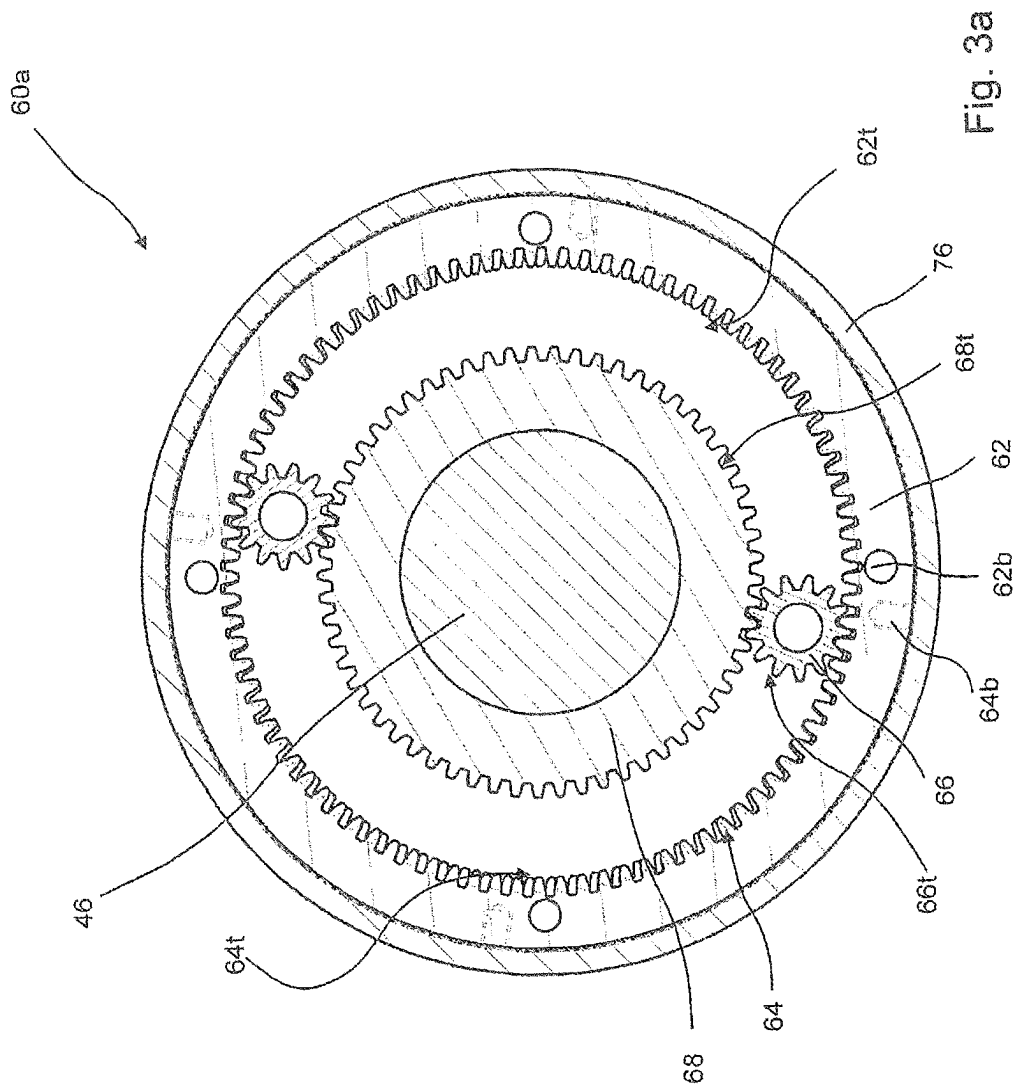
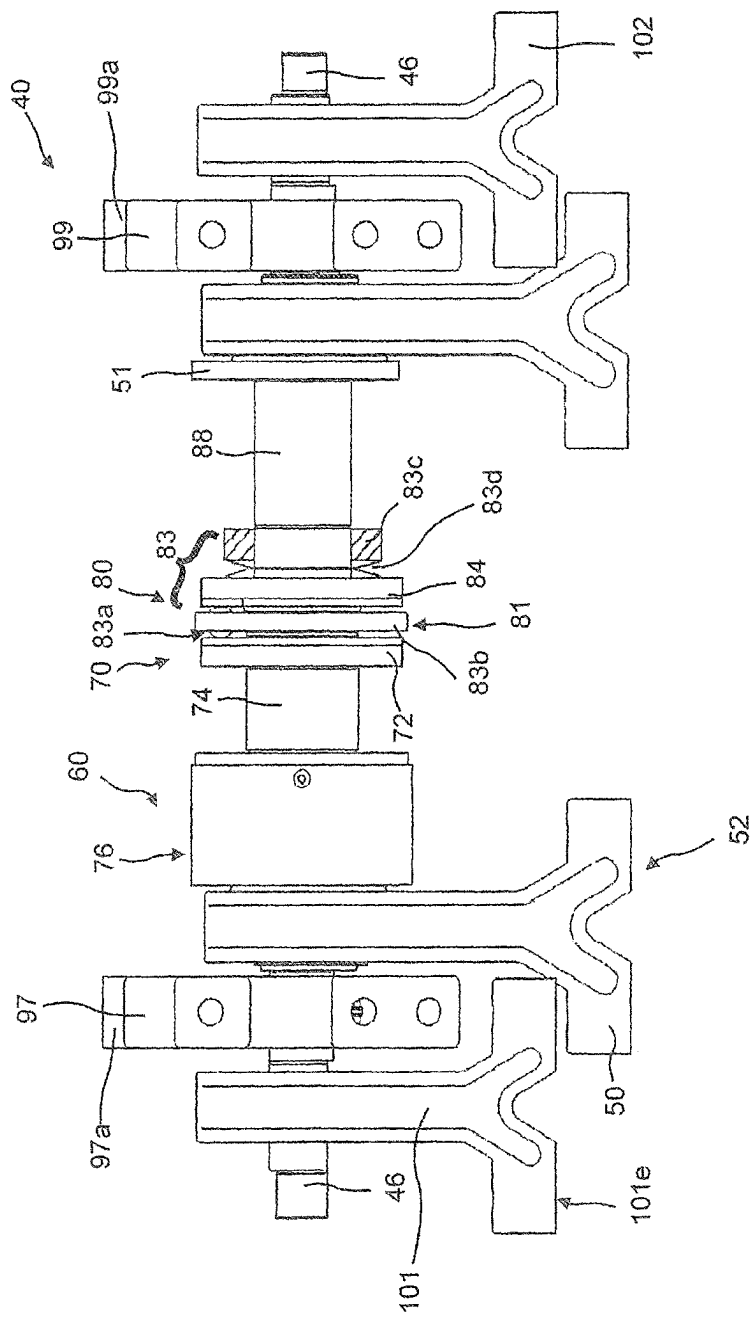
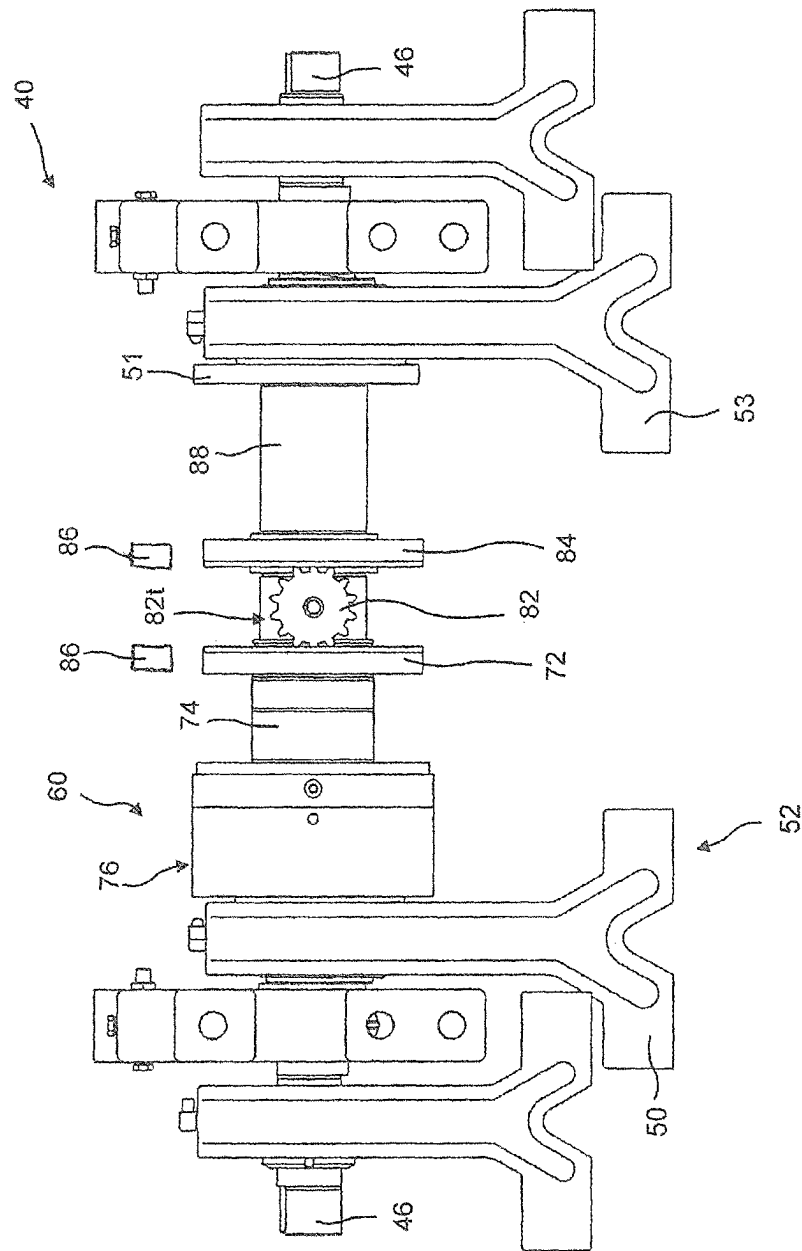


Fig 3







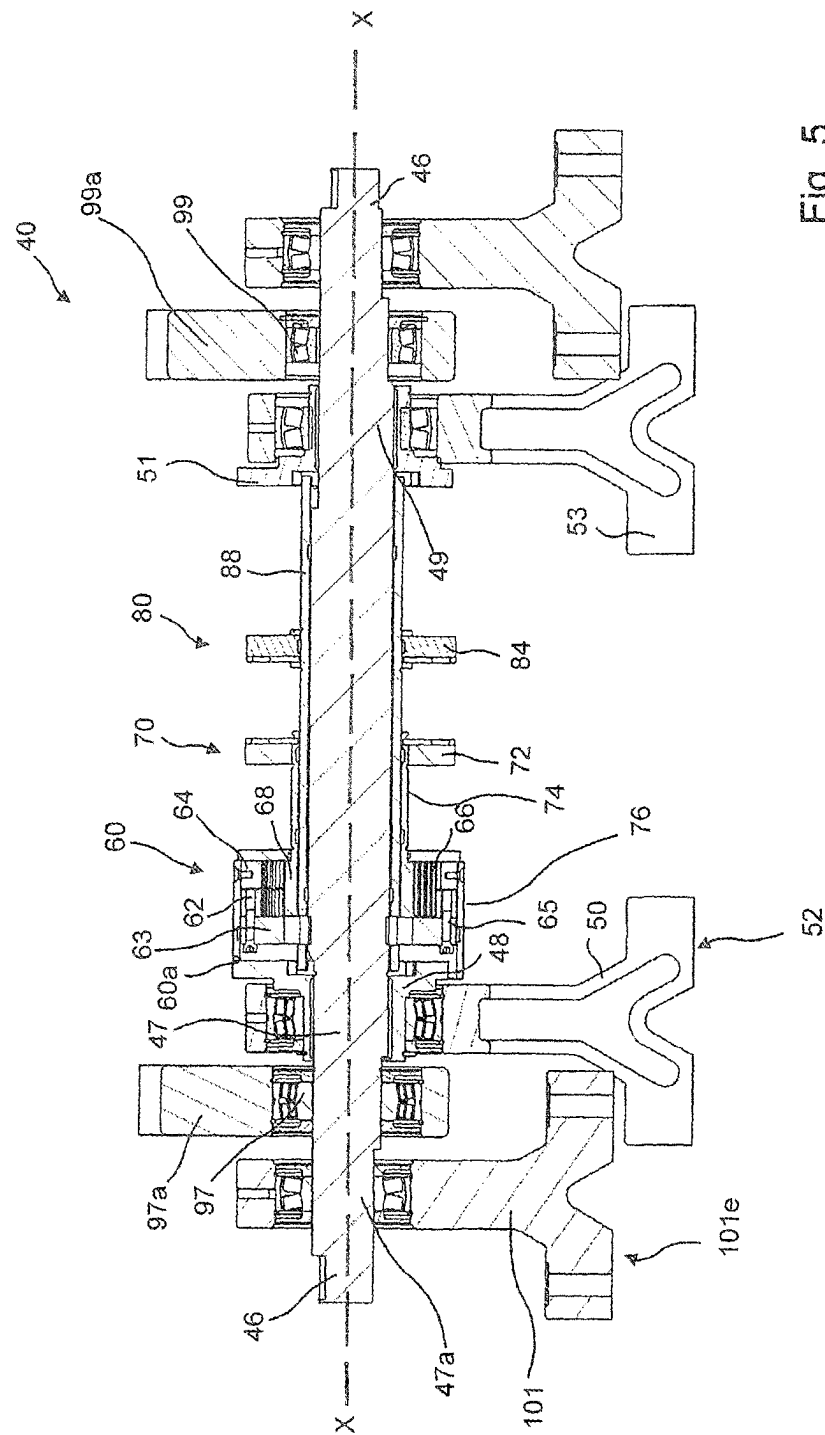


Fig 5

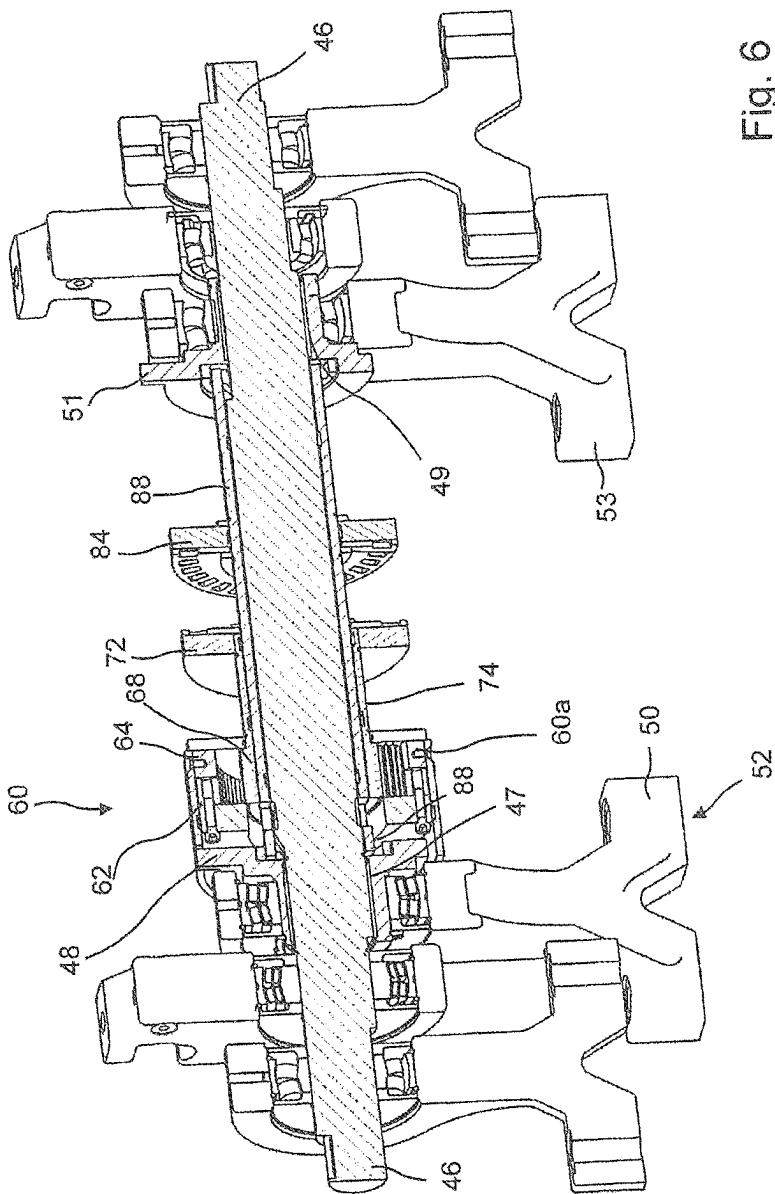


Fig. 6

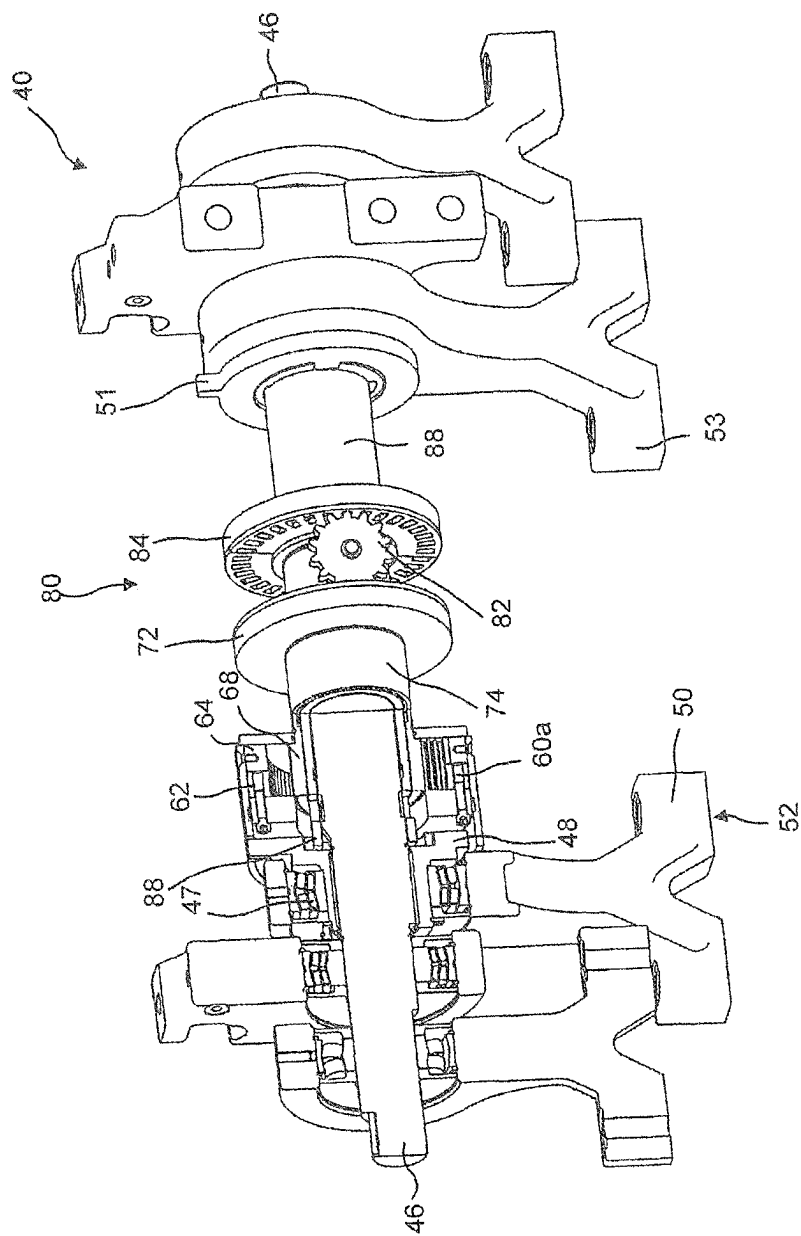
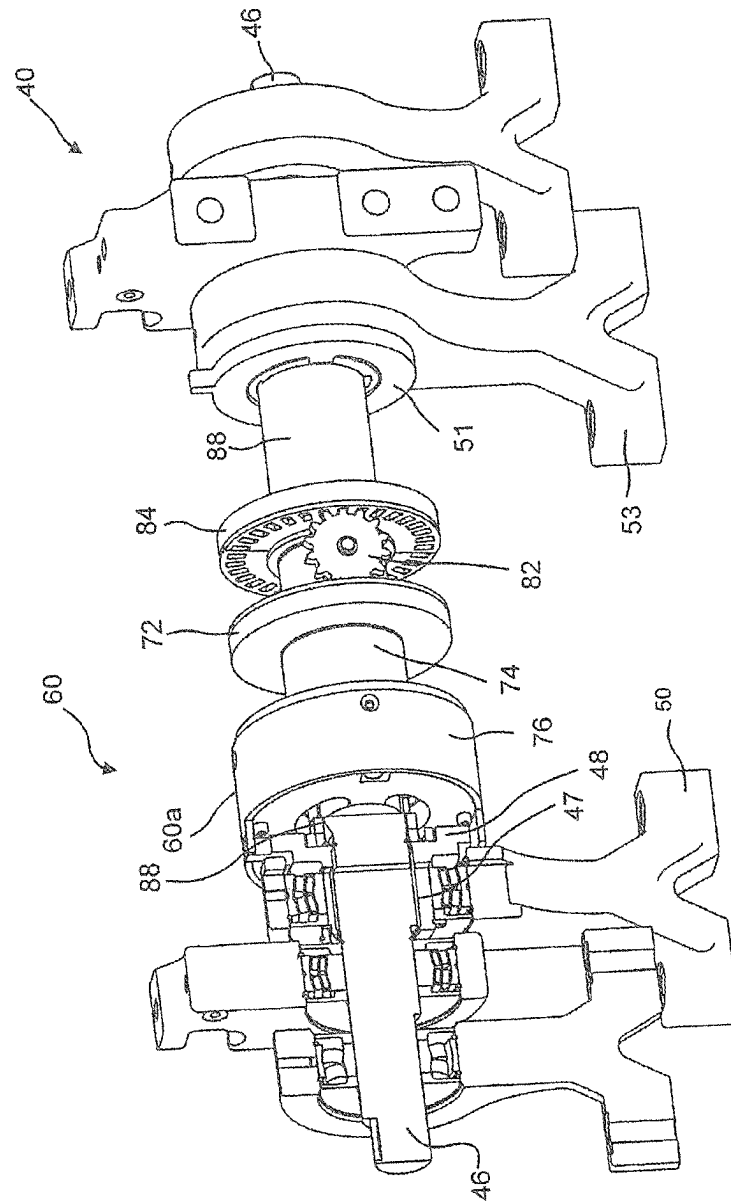


Fig. 7




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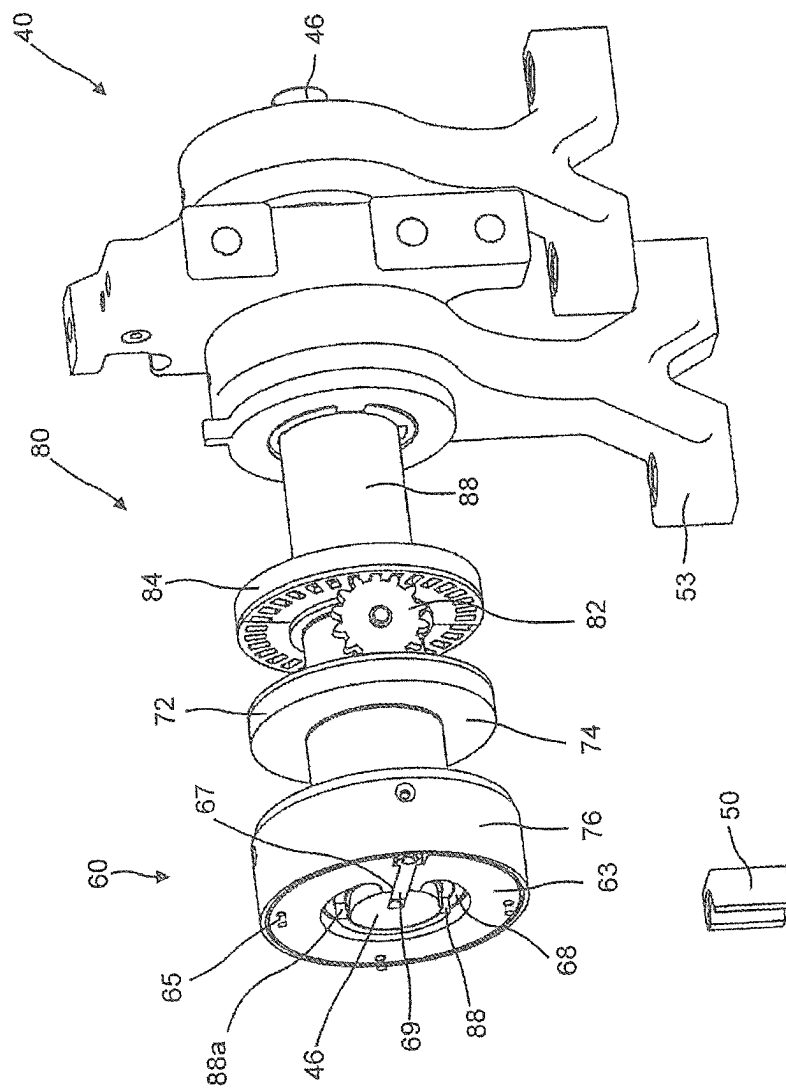


Fig. 9

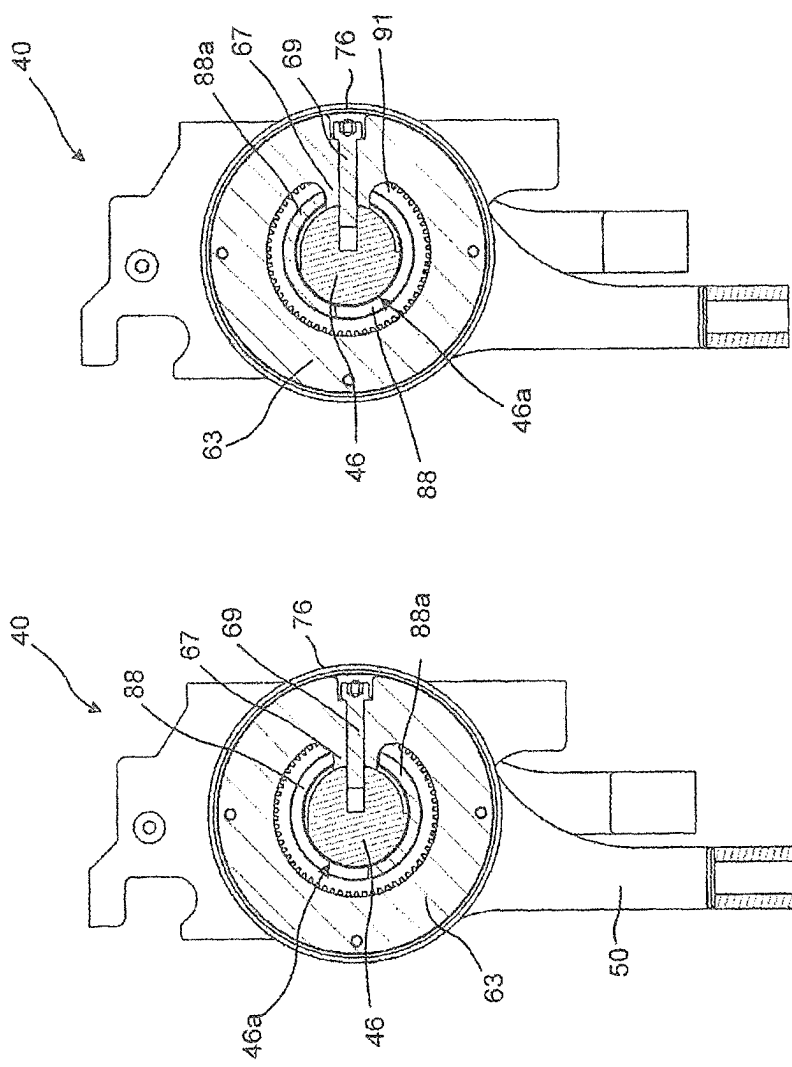


Fig. 10b

Fig. 10a

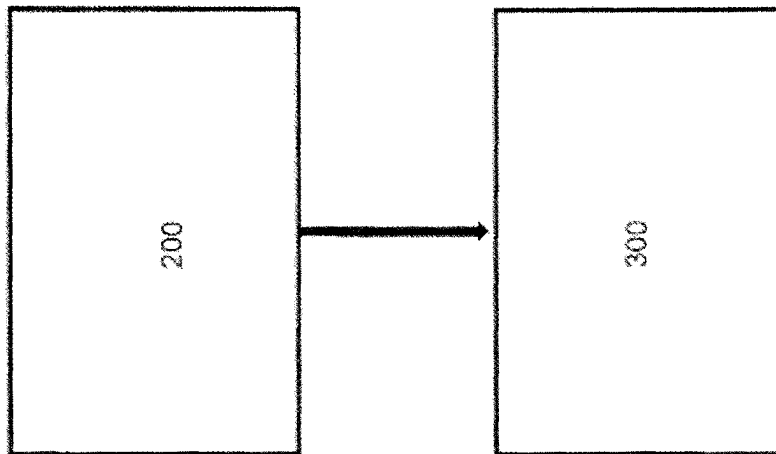


Fig. 11

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**TAMPER DEVICE FOR A SCREED OF A
WORKING MACHINE AND A METHOD FOR
ADJUSTING A STROKE OF A TAMPER
DEVICE FOR A SCREED OF A WORKING
MACHINE**

CROSS REFERENCE TO RELATED
APPLICATIONS

This application is a 35 U.S.C. § 371 national stage application of PCT International Application No. PCT/EP2020/051908 filed on Jan. 27, 2020, the disclosure and content of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The invention relates to a tamper device for a screed of a working machine and a method for adjusting a stroke of a tamper device for a screed of a working machine.

The invention is applicable on working machines within the fields of industrial construction machines or construction equipment, in particular road pavers or asphalt finishers. Although the invention will be described with respect to a road paver, the invention is not restricted to this particular machine, but may also be used in other pavers or other working machines.

BACKGROUND

A tamper device for a screed of a road paver is known from US 2011/0123270 A1. The tamper device according to US 2011/0123270 A1 comprises an eccentric shaft comprising an eccentric section and an eccentric bushing arranged thereon which is rotatable mounted in a connecting rod driving a tamper bar. The stroke of the tamper bar is adjustable by a relative rotation between the eccentric shaft and the eccentric bushing. A tappet and a preselection area with two tappet stop positions defining two different tamper bar strokes are functionally provided between the eccentric shaft and the eccentric bushing, so that the tappet can be adjusted without tools to each tappet stop position by a reversal of the direction of rotation of the eccentric shaft to switch between the two strokes.

SUMMARY

An object of the invention is to provide an improved tamper device for a screed of a working machine and an improved method for adjusting a stroke of a tamper device for a screed of a working machine, in particular to provide a tamper device for a screed of a working machine and a method for adjusting a stroke of a tamper device for a screed of a working machine which provide for an adjustment of the stroke while the tamper device is running.

According to a first aspect of the invention, the object is achieved by a tamper device according to claim 1. The tamper device for a screed of a working machine, in particular a paver, comprises: a rotatable driveable tamper shaft comprising an eccentric section, an eccentric bushing mounted on the eccentric section, and a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke. The stroke is adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section. The tamper device is characterized by an inner-toothed hollow wheel gear train for providing the rotational

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adjustment. The gear train is connected to the tamper shaft, to the eccentric bushing, and to a drive force receiving element. The drive force receiving element is configured to be able to receive a drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating.

In the present application an inner-toothed hollow wheel gear train is a gear train which comprises an inner-toothed hollow wheel. In the present application the term inner-toothed hollow wheel means that there is a hollow wheel which comprises teeth on its inside surface. Such an inner-toothed hollow wheel can for example be a ring gear of a planetary gear or a circular spline of a strain wave gear. The latter also known as an harmonic gear.

The present invention comprises the perception that with the above-referenced known tamper device the stroke of the connecting rod driving the tamper bar can only be switched between two strokes at two preselected positions. This is a serious drawback since the local position of the screed might require an adjustment of the stroke which adjustment is not matching to the preselected positions. Since these two preselected positions can only be reached by a reversal of the direction of rotation of the eccentric shaft, it is necessary to stop the tamper to switch between the two strokes. This is a serious drawback since a stop of the tamper leads to a stop of the paver.

By the provision of a tamper device which comprises a rotatable driveable tamper shaft comprising an eccentric section, an eccentric bushing mounted on the eccentric section, and a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section, and which tamper device is characterized by an inner-toothed hollow wheel gear train for providing the rotational adjustment, the gear train being connected to the tamper shaft, to the eccentric bushing, and to a drive force receiving element being configured to be able to receive a drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating, it is in particular provided the advantage of an adjustability of the stroke while the tamper device is running.

An advantage of the tamper device of the present invention is that the stroke of the connecting rod driving a tamper bar can be adjusted individually and arbitrarily. This is a strong advantage since an individual local position of the screed might require an individual adjustment of the stroke. If for example a road paver uses more than one screed section, for example a basic screed and optionally screed extensions that can be extended at the basic screed for changing the working width of the road paver, each of these components of the screeds may comprise its own tamper device, so that the present invention enables an individual adjustment of the stroke for each screed section. Due to the inventive use of an inner-toothed hollow wheel gear train in the inventive tamper device an adjustment of the stroke can be provided during rotation of the tamper shaft. It is therefore in particular not necessary to stop the tamper to adjust the stroke. This is a strong advantage since therefore a costly stop of the work of the working machine, for example a stop of a paver, can be avoided.

According to one embodiment, the inner-toothed hollow wheel gear train comprises a planetary gear. The use of a planetary gear is a reliable and cost-efficient implementation of the present invention.

According to a further embodiment, the planetary gear is a two-stage planetary gear and comprises a shaft side ring

gear being connected to the tamper shaft in a torque-proof manner, a bushing side ring gear being connected to the eccentric bushing in a torque-proof manner, a common planet gear connecting the shaft side ring gear with the bushing side ring gear, and a common sun gear being connected to the drive force receiving element in a torque-proof manner. By connecting a shaft side ring gear being an inner-toothed hollow wheel with the tamper shaft in a torque proof manner and connecting a bushing side ring gear being another inner-toothed hollow wheel with the eccentric bushing in a torque-proof manner, and using the common sun gear of such two-stage planetary gear as the drive force receiving element a simple, reliable and cost-efficient configuration of the inventive gear train is provided for driving the gear train when the tamper shaft is running.

A further embodiment comprises a housing for the shaft side ring gear, the bushing side ring gear, the planet gear, and the sun gear, the housing being connected to the eccentric bushing in a torque-proof manner, the bushing side ring gear being connected to the housing in a torque-proof manner, whereas the shaft side ring gear, the planet gear, and the sun gear being rotatable relative to the housing. By providing a housing for the gears of the planetary gear and by connecting the housing to the eccentric bushing of the tamper shaft in a torque-proof manner it is provided a simple and reliable protection of the gears of the planetary gear in a possibly rough environment when the inventive tamper device is for example used on a road paver, while simultaneously the housing is the second stage or the output side of such planetary gear.

According to a further embodiment the shaft side ring gear comprises less teeth than the bushing side ring gear, preferably a ratio of a number of teeth of the shaft side ring gear to a number of teeth of the bushing side ring gear being about 1:30 to about 1:250. Such range of ratios have been found as being particularly useful for the requirements of road pavers.

According to a further embodiment the shaft side ring gear comprises more teeth than the bushing side ring gear, preferably a ratio of a number of teeth of the shaft side ring gear to a number of teeth of the bushing side ring gear being about 30:1 to about 250:1. Such range of ratios have been found as being particularly useful for the requirements of road pavers.

According to a further embodiment the shaft side ring gear comprises 87 teeth and the bushing side ring gear comprises 89 teeth, the planet gear comprises 13 teeth, and the sun gear comprises 62 teeth. Such range of ratios have been found as being particularly useful for the requirements of road pavers.

According to a further embodiment the drive force receiving element comprises a brake force receiving element being configured to be able to receive a brake force for driving the inner toothed hollow wheel gear train when the tamper shaft is rotating. This embodiment of the present invention is a particularly advantageous implementation of the present invention. This is because the present invention comprises the perception that by using an inner-toothed hollow wheel gear train it is not only possible to provide the rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section while the tamper shaft is rotating or running but it is additionally possible to use the rotational energy of the rotating tamper shaft for driving the inventive gear train by providing a brake force receiving element receiving a brake force for driving the gear train. By for example connecting the brake force receiving element with the input side of the inventive gear

train as described herein the brake force receiving element may simply rotate together with the whole gear train which itself is rotating together with the tamper shaft due to its connection to the tamper shaft. Only if an adjustment of the relative rotational positioning between the eccentric bushing and the eccentric section for an adjustment of the stroke is desired, a brake force is provided to the brake force receiving element connected with the input side of the inventive gear train to drive the inventive gear train to thereby provide the rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section.

According to a further embodiment the brake force receiving element is configured to be able to receive a brake force by comprising an adjustment wheel sitting on the tamper shaft, the adjustment wheel being rotatable relative to the tamper shaft, the adjustment wheel being connected to the inner-toothed hollow wheel gear train in a torque-proof manner, whereby the adjustment wheel is being configured to drive the eccentric bushing in a direction against a direction of rotation of the tamper shaft, in case the adjustment wheel is receiving a brake force when the tamper shaft is rotating. By using an adjustment wheel sitting on the tamper shaft a simple and cost-efficient implementation of a brake force receiving element is provided.

According to a further embodiment the adjustment wheel is connected to the inner-toothed hollow wheel gear train in a torque-proof manner via a connecting tube surrounding the tamper shaft. By connecting the adjustment wheel to the inventive gear train via a tube surrounding the tamper shaft a reliable and cost efficient implementation of a torque-proofed connection between the inventive adjustment wheel and the inventive gear train is provided.

According to a further embodiment, in case the gear train comprises the planetary gear as described herein, the adjustment wheel is connected to the sun gear in a torque-proof manner. By connecting the adjustment wheel with the sun gear of a planetary gear as described herein in a torque-proof manner a simple and cost-efficient implementation of a connection between the adjustment and a planetary gear used as the inventive gear train is provided since the sun gear is the inner most part of a planetary gear so that it is in particular simple to for example use a connecting tube as mentioned above to connect the adjustment wheel and the sun gear.

A further embodiment comprises a further brake force receiving element being connected with the brake force receiving element via a gear unit, the further brake force receiving element being configured to be able to receive a brake force for driving the gear unit by comprising a further adjustment wheel sitting on the tamper shaft, the further adjustment wheel being rotatable relative to the tamper shaft, whereby the further adjustment wheel is being configured to drive the adjustment wheel in a direction of rotation of the tamper shaft, in case the further adjustment wheel is receiving a brake force when the tamper shaft is rotating. By providing a further brake force receiving element in addition to the brake force receiving element, and by connecting the further brake force receiving element with the brake force receiving element via a gear unit, it is advantageously possible by for example using a further adjustment wheel for the further brake force received element to provide a brake force to the further adjustment wheel which brake force is then translated by the gear unit being connected to the adjustment wheel so that the adjustment wheel can be rotated in the same direction as the rotation direction of the tamper shaft by providing a brake force to the further adjustment wheel.

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Accordingly, due to this particularly advantageous embodiment of the present invention it is possible by using two adjustment wheels, the adjustment wheel and the further adjustment wheel, connected via a gear unit, to either brake on the adjustment wheel to drive the eccentric bushing in a direction against a direction of rotation of the tamper shaft, or to brake on the further adjustment wheel so that its rotation is slowed down with respect to the rotation of the tamper shaft, this slowing down of the further adjustment wheel being translated by the gear unit into an acceleration of the adjustment wheel which drives the eccentric bushing in a direction of rotation of the tamper shaft. With other words: By the present embodiment the direction of rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section can be chosen by choosing either the adjustment wheel or the further adjustment wheel to be braked on. Thereby a necessity to provide a rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section of more than 180° is advantageously avoided. Accordingly, an adjustment on the basis of this embodiment of the present invention substantially enhances the velocity of the rotational adjustment.

According to a further embodiment the gear unit comprises a cog wheel and/or a friction gear. Hereby reliable and cost-efficient implementations of the present invention are provided.

A further embodiment comprises a mechanical brake and/or an eddy current brake for providing a brake force to the brake force receiving element and/or the further brake force receiving element. Hereby reliable and cost-efficient implementations of the present invention are provided.

A further embodiment comprises a tamper bar being mounted at an end of the connecting rod.

According to a further embodiment a shaft eccentricity of the eccentric section and a bushing eccentricity of the eccentric bushing are configured so that the stroke is adjustable between a predefined minimum and a predefined maximum. Preferably, if the predefined minimum is zero, this embodiment provides the possibility to not only arbitrarily adjust the stroke between a predefined non-zero minimum and a predefined maximum but to adjust the stroke even to zero. Such a possibility may be useful in particular if a screed used two parallel tamper bars, for example one tamper bar being mounted at the ends of two connecting rods having an adjustable stroke according to the present invention, and another tamper bar located downstream of the afore-mentioned tamper bar having a fixed stroke. By this embodiment it would then be possible to adjust the stroke of the upstream tamper bar to a predefined minimum. The minimum may be zero if for example the impact of the stroke of the second tamper bar would be sufficient for a certain working condition of a respective road paver.

According to a further embodiment the tamper shaft comprises a further eccentric section with a further eccentric bushing mounted on the further eccentric section, and a further connecting rod being rotatable mounted on the further eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the further eccentric bushing and the further eccentric section, wherein the eccentric bushing and the further eccentric bushing are connected in a torque-proof manner by an elongated tube surrounding the tamper shaft between the eccentric bushing and the further eccentric bushing. By this embodiment it is possible to in parallel adjust the stroke of the connecting rod and the further connecting rod by con-

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necting the eccentric bushings of each of the connecting rods so that a relative rotational positioning of the eccentric bushing according to the present invention causes a parallel additional positioning of the further eccentric bushing.

According to a further embodiment the elongated tube is mounted at each bushing with lateral play to compensate a lateral movement of the bushings due to the eccentricity of the respective eccentric sections on the tamper shaft when the stroke being adjusted by an rotational adjustment of a relative rotational positioning between the eccentric bushings and the eccentric sections. By mounting the elongated tube connected the eccentric bushing with the further eccentric bushing at each bushing with lateral play this embodiment advantageously compensates a lateral movement of the bushings due to the eccentricity of the respective eccentric sections on the tamper shaft when the stroke being adjusted by an rotational adjustment of a relative rotational positioning between the eccentric bushings and the eccentric sections of the tamper shaft.

A further embodiment comprises an electric motor, and the drive force receiving element is configured to be driven by the electric motor. By providing, an electric motor and by configuring the drive force receiving element to be driven by the electric motor, for example by connecting the output side of the electric motor with the drive force receiving element a well-known, reliable and cost-efficient implementation for driving the inventive gear train is provided. Additionally, by using a motor, the drive force receiving element can receive a drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating, as well as when the tamper shaft is standing still.

According to a further embodiment the inner-toothed hollow wheel gear train comprises a strain wave gear. The use of a strain wave gear, also known as a harmonic gear or harmonic drive, provides the advantages of nearly no backlash, enhanced compactness, reduced weight, high gear ratios, reconfigurable ratios within the same housing, enhance resolution and excellent repeatability when repositioning initial loads, and a high torque capability. In particular, the use of a strain wave gear provides high gear reduction ratios in a small volume. As an example, it is possible to realize a gear reduction ratio from 30:1 up to 320:1 in the same space in which a planetary gear typically produces a reduction ratio of 10:1.

According to a further embodiment the strain wave gear comprises a circular spline being connected to the tamper shaft in a torque-proof manner, a flex spline being connected to the eccentric bushing in a torque-proof manner, and a drive element for moving the flex spline and being connected to the drive force receiving element in a torque-proof manner. By connecting the circular spline with the tamper shaft and therefore with the eccentric section of the tamper shaft in a torque-proof manner a simple and reliable implementation of a strain wave gear as the gear train according to the present invention is provided. In an alternative embodiment the circular spline being connected to the eccentric bushing in a torque-proof manner while the flex spline is connected to the tamper shaft and therefore to the eccentric section of the tamper shaft in a torque-proof manner.

In case the drive force receiving element of this further embodiment comprises a brake force receiving element being configured to be able to receive a brake force for driving the strain wave gear when the tamper shaft is rotating, the brake force receiving element may be configured to be able to receive a brake force by comprising an adjustment wheel sitting on the tamper shaft, the adjustment

wheel being rotatable relative to the tamper shaft and being connected to the flex spline in a torque-proof manner, preferably via a connecting tube surrounding the tamper shaft, whereby the adjustment wheel is being configured to drive the eccentric bushing in a direction against a direction of rotation of the tamper shaft, in case the adjustment wheel is receiving a brake force when the tamper shaft is rotating. Hereby a reliable and cost-efficient implementation of the present invention is provided. As an alternative implementation, the adjustment wheel may be connected to a circular spline in a torque-proof manner of a strain wave gear as described herein.

The present invention also relates to a screed of a working machine, in particular a paver, comprising a tamper device as described herein.

The present invention also relates to a working machine, in particular a road paver, comprising a screed as described herein.

According to a second aspect of the present invention, the object is achieved by a method for adjusting a stroke of a tamper device for a screed of a working machine according to claim 19. The tamper device comprising a rotatable driveable tamper shaft comprising an eccentric section, an eccentric bushing mounted on the eccentric section, and a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section. The method comprises the steps of providing the rotational adjustment by driving an inner-toothed hollow wheel gear train by providing a drive force to a drive force receiving element. The drive force receiving element is connected to the inner-toothed hollow wheel gear train. The drive force receiving element is configured to be able to receive the drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating. The inner-toothed hollow wheel gear train is also connected to the tamper shaft and to the eccentric bushing.

By the provision of a method for adjusting a stroke of a tamper device for a screed of a working machine, the tamper device comprising: a rotatable driveable tamper shaft comprising an eccentric section, an eccentric bushing mounted on the eccentric section, and a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section, which method is characterized by the steps of providing the rotational adjustment by driving an inner-toothed hollow wheel gear train by providing a drive force to a drive force receiving element connected to the inner-toothed hollow wheel gear train and being configured to be able to receive the drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating, the inner-toothed hollow wheel gear train also being connected to the tamper shaft and to the eccentric bushing, it is in particular provided the advantage of an adjustability of the stroke while the tamper device is running.

An advantage of the method of the present invention is that the stroke of the connecting rod driving a tamper bar can be adjusted individually. This is a strong advantage since an individual local position of the screed might require an individual adjustment of the stroke. Due to the inventive use of an inner-toothed hollow wheel gear train in the inventive tamper device, an adjustment of the stroke can be provided during rotation of the tamper shaft. It is therefore in par-

ticular not necessary to stop the tamper to adjust the stroke. This is a strong advantage since therefore a costly stop of the work of the working machine, for example a stop of a paver, can be avoided.

According to one embodiment, the method comprises the further step of measuring an angle of the relative rotational positioning between the eccentric bushing and the eccentric section. Hereby an exact information about the relative rotational positioning between the eccentric bushing and the eccentric section can be retrieved.

According to a further embodiment, the method comprises the further step of using the measured angle of the relative rotational positioning to define the amount of rotational adjustment of the relative rotational positioning between the eccentric bushing and the eccentric section. Hereby a precise adjustment of the relative rotational positioning between the eccentric bushing and the eccentric section and thereby a precise determination of the stroke can be achieved. This possibility is particularly advantageous if such method is for example used in combination with a measurement of a laid material thickness of the paved material and/or of a measurement of compaction of the paved material so that the stroke can immediately be adjusted on the basis of the measured parameters of material thickness and/or compaction and the actual stroke derived from the actually measured angle of the relative rotational positioning.

The present invention also relates to a computer program comprising program code means for performing the steps of the method for adjusting a stroke of a tamper device for a screed of a working machine as described herein, when said program is run on at least one computer.

The present invention also relates to a computer readable medium carrying a computer program comprising program code means for performing the steps of the method for adjusting a stroke of a tamper device for a screed of a working machine as described herein, when said program product is run on at least one computer.

The present invention also relates to a control unit for controlling a tamper device for a screed of a working machine, in particular a paver, the control unit being configured to perform the steps of the method for adjusting a stroke of a tamper device for a screed of a working machine as described herein.

According to one embodiment, the control unit comprises a mechanical and/or electrical sensor for measuring an angle of the relative rotational positioning between the eccentric bushing and the eccentric section.

The present invention also relates to a working machine, in particular a road paver, comprising a control unit as described herein.

Further advantages and advantageous features of the invention are disclosed in the following description and in the dependent claims.

BRIEF DESCRIPTION OF THE DRAWINGS

With reference to the appended drawings, below follows a more detailed description of embodiments of the invention cited as examples. In the drawings and the following detailed description of the drawings, identical elements or elements with the same function are indicated with the same reference numeral.

In the drawings:

FIG. 1 is an illustration of a schematically side view of a road paver;

FIG. 2 is a schematic illustration, partly in cross-section, of a screed of a working machine;

FIGS. 2a and 2b are schematic and simplified illustrations of two exemplary working positions of a downstream connecting rod according to FIG. 2;

FIGS. 2c and 2d are schematic and simplified illustrations of two exemplary working positions of an upstream connecting rod according to FIG. 2;

FIGS. 2e and 2f are schematic and simplified cross-sections through the upstream connecting rod along plane A-A shown in FIGS. 2c and 2d, respectively;

FIG. 3 is a schematic illustration of an exemplary embodiment of the invention;

FIG. 3a is a cross section through the shaft side ring gear along plane B-B in FIG. 3;

FIG. 4a is an illustration of another embodiment of the present invention;

FIG. 4b is an illustration of another embodiment according to the present invention;

FIG. 5 shows a cross-sectional view of the embodiment according to FIG. 4b;

FIG. 6 is a perspective view of FIG. 5;

FIG. 7 corresponds to FIG. 6 but does only show a partial cross-section;

FIG. 8 corresponds to FIG. 7 with a reduced cross-sectional part;

FIG. 9 corresponds to FIG. 8 but shows part of FIG. 8 cut away;

FIGS. 10a and 10b are side views from left to right in FIG. 9 on the left hand side of the device of FIG. 9; and

FIG. 11 is a schematic illustration of an inventive method.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS OF THE INVENTION

FIG. 1 is an illustration of a schematically side view of a road paver 10. The road paver 10 comprises a frame 12 with a set of ground-engaging elements 14 such as tracks or wheels coupled with the frame 12. The elements 14 may be driven by an engine inside the frame in a conventional manner. The engine may further drive an associated generator in a conventional manner to drive a screed 18 of the road paver 10. The screed 18 is attached at the rear end of the road paver 10 to spread a compact paving material into a mat 20 as shown in FIG. 2. The road paver 10 additionally comprises a hopper 26 for storing paving material, and a conveyer system to move the paving material from the hopper 26 to a deflector plate 27 in front of the screed 18.

The screed 18 is pivotally connected behind the road paver 10 by a pair of tow arms 28 that extend on each side of the frame of the road paver 10 between the frame 12 and the screed 18. The tow arms 28 are connected to frame 12 in a pivotable manner so that the position and orientation of the screed 18 relative to the frame 12 and to the surface being paved may be adjusted by raising or lowering the tow arm actuators to control the thickness of the paving material deposited by the road paver 10 below a base plate 30 of the screed 18.

FIG. 2 is a schematic illustration, partly in cross section, of a screed 18 of a working machine, of for example a road paver 10 of FIG. 1, comprising a tamper device 40 as described herein. The construction and function of the screed 18 of FIG. 2 is substantially the same as described above with respect to the screed of the road paver 10 of FIG. 1. Such screed 18 may have any configurations known from the prior art. In particular, screed 18 of FIG. 1 may be a single or a multiple sections screed.

Screed 18 may include a screed extension provided behind and adjacent to each of the left and right main screed sections. The screed extensions may be slidable moveable laterally between retracted and extended positions such that varying width of paving material 20 can be laid. The lateral movement of the extensions of the screed 18 may be driven by respective screed width actuators. Although not shown in FIG. 2, each of a main screed 18 and possible extendable extension screeds and other possible screed broadening parts may be provided with at least one tamper device 40 as exemplary explained herein for the screed 18 of FIG. 2.

As shown in FIGS. 1-2, tamper device 40 comprises an upstream tamper bar 42 and a downstream tamper bar 44. The upstream tamper bar 42 is closer to the deflector plate 27, while the downstream tamper bar 44 is closer to the base plate 30 of the screed 18.

Upstream tamper bar 42 is mounted at a lower end 52 of a connecting rod 50. Downstream tamper bar 44 is mounted at a lower end 101e of a downstream connecting rod 101. Both connecting rods 50 and 101 are mounted on a rotatable drivable tamper shaft 46 of screed 18.

FIGS. 2a and 2b are schematic and simplified illustrations of two exemplary working positions of a downstream connecting rod 101 according to FIG. 2 in cross section. FIG. 2a shows a first exemplary working position of downstream connecting rod 101 and FIG. 2b shows a second exemplary working position of downstream connecting rod 101. FIGS. 2a and 2b serve to illustrate the structure and the working principle of how tamper shaft 46 lifts and lowers connecting rod 101 as tamper shaft 46 rotates.

As shown in FIGS. 2a and 2b, tamper shaft 46 is rotatable drivable around its longitudinal axis X as indicated by arrow 46a. Tamper shaft 46 is rotatable mounted in bearings 97 and 99 which are located in respective housing structures 97a and 99a which are part of screed 18. Tamper shaft 46 comprises an eccentric section 47a, i.e. a section having a central longitudinal axis laterally shifted with respect to the longitudinal axis X of rotation of tamper shaft 46. Tamper shaft 46 is rotatable in a bearing 101a of connecting rod 101. Connecting rod 101 is mounted with bearing 101a on the eccentric section 47. Mounted at an end 101e of connecting rod 101 is the tamper bar 44 not shown in FIGS. 2a and 2b.

Tamper device 40 according to FIGS. 2a and 2b operates as follows: Rotation of tamper shaft 46 around its longitudinal axis X according to arrow 46a also rotates eccentric section 47a of tamper shaft 46. The rotation of eccentric section 47a in bearing 101a of connecting rod 101 causes connecting rod 101 moving up and down as indicated by double arrow 101b, in FIG. 2a, eccentric section 47 is depicted as having reached its lowermost position so that also end 101e of connecting rod 101 has reached its lowermost position. Further rotating tamper shaft 46 according to arrow 46a with an amount of 180° moves eccentric section 47 in an opposite and uppermost position as illustrated in FIG. 2b.

This rotation of section 47 into the uppermost position also causes connecting rod 101 to reach its uppermost position according to FIG. 2b. The difference between the lowermost position according to FIG. 2a and the uppermost position of connecting rod 101 is a stroke S of tamper bar 44. Stroke S is indicated in FIG. 2b by a curly bracket showing the distance S between a dashed line indicative of the lowermost position according to FIG. 2a, and end 101e of rod 101. The eccentricity of the eccentric section 47 determines the stroke S of connecting rod 101 and tamper bar 44, respectively.

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Continuing this rotation according to arrow **46a** causes an up and down movement of connecting rod **101** according to arrow **101b**. Tamper bar **44** mounted at end **101e** therefore conducts the same up and down movement according to arrow **101b**.

The tamper shaft **46** with eccentric section **47** may be rotated according to arrow **46a** at a certain rotational speed. The rotational speed determines the frequency of the vertical up and down movements **101b**. The rotational speed of a rotation **46a** of the tamper shaft **46** and thus the operating frequency of connecting rod **101** and tamper bar **44** is preferably set to provide a desired compaction result at a predefined paving speed of the paver **10** for the used road construction material.

FIGS. **2c** and **2d** are schematic and simplified cross sections to illustrate an adjustment ΔS of the stroke **S** by a rotation of eccentric bushing **48** by showing two exemplary adjustments of a working position of an upstream connecting rod **50** according to FIG. **2**. FIGS. **2e** and **2f** are schematic and simplified cross-sections through the upstream connecting rod **50** along plane A-A as shown in FIGS. **2c** and **2d**, respectively, to further clarify and facilitate understanding of the adjustment of the stroke **S** by a rotation of eccentric bushing **48**.

As shown in FIGS. **2c** to **2f**, similar to downstream rod **101**, also upstream rod **50** sits on an eccentric section **47** of tamper shaft **46** so that generally also rod **50** may move up and down with a stroke **S** while shaft **46** is rotating in rod **50**. But the stroke **S** of rod **50** may be adjusted, and FIGS. **2c** to **2f** serve to illustrate the structure and the working principle of how to adjust a stroke **S** of the upstream connecting rod **50** and thereby of the upstream tamper bar **42** of FIG. **2**.

As shown in FIGS. **2c** to **2f**, the difference between the mounting of downstream rod **101** on shaft **46** and the mounting of upstream rod **50** on shaft **46** is that the bearing **101a** of downstream rod **101** is mounted directly on shaft **46** while a bearing **50a** of upstream rod **50** is not directly mounted on shaft **46**. This is because an eccentric bushing **48** is located between bearing **50a** of upstream rod **50** and shaft **46**. Eccentric bushing **48** is located on the eccentric section **47** of shaft **46**. Eccentric bushing **48** is rotatable in bearing **50a** of rod **50**. Additionally, eccentric bushing **48** is rotatable with respect to the eccentric section **47**, independent of a rotational position of eccentric section **47** of shaft **46**.

To illustrate the possibility of adjusting the stroke of connecting rod **50**, and thereby of tamper bar **52** mounted at an end **52** of connecting rod **50**, FIGS. **2c** and **2e** show one exemplary adjusted working position of rotatable eccentric bushing **48** on the eccentric section **47** on tamper shaft **46**, and FIGS. **2d** and **2f** show another exemplary adjusted working position of rotatable eccentric bushing **48** on the eccentric section **47** on tamper shaft **46**. To facilitate understanding of the adjustment of the stroke by rotating eccentric bushing **48**, FIGS. **2c** to **2f** show the two exemplary adjusted positions of eccentric bushing **48** for an unchanged rotational position of tamper shaft **46**.

An adjustment of the stroke of the connecting rod **50** works as follows: As shown in FIGS. **2c** and **2e**, a rotational position of eccentric bushing **48** has been rotated relative to the eccentric section **47** on shaft **46** to move an eccentricity of eccentric bushing **48** into a position where it faces an end **52** of connecting rod **50**.

In FIGS. **2d** and **2f**, eccentric bushing **48** has been rotated relative to the eccentric section **47** on shaft **46** to move the eccentricity of eccentric bushing **48** where it faces away from end **52** of connecting rod **50**, thereby adjusting the

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position of end **52** of connecting rod **50** by an amount of ΔS as indicated by the curly bracket in FIGS. **2d** and **2f**.

Since each of FIGS. **2c** to **2f** show the eccentric section **47** on tamper shaft **46** in the same rotational position with the eccentricity of eccentric section **47** in the uppermost position, the illustrated adjustment ΔS of the stroke of rod **50** is an adjustment by a distance **LS** of the uppermost position of the stroke **S** of rod **50** and therefore of the uppermost position of the corresponding stroke **S** of tamper bar **42** of tamper device **40**.

While FIGS. **2a** to **2d** show tamper shaft **46** being mounted in two bearings **97** and **99** being located in two corresponding housing structures **97a** and **99a**, a person of ordinary skill in the art will appreciate that tamper shaft **46** may be mounted in additional bearings and housing structures or may be mounted in only one of bearings **97** or **99** in one corresponding housing structure **97a** or **99a**. Additionally, a person of ordinary skill in the art will appreciate that the depicted distance between connecting rods **101** and **50**, respectively, and an adjacent housing structure **97a** or **99a** may be reduced or enhanced as desired. It may for example be enhanced to mount one or more additional connecting rods on tamper shaft **46** between connecting rods **101** and **50**, respectively, and an adjacent housing structure **97a** or **99a**, if desired.

While in FIGS. **2a** to **2f** eccentric sections **47a** and **47** on shaft **46** are depicted as having a larger diameter than adjacent sections of shaft **46**, a person of ordinary skill in the art will appreciate that the eccentric sections **47a** and **47** may also function to provide eccentricity to shaft **46** when having a smaller diameter than adjacent sections of shaft **46**.

FIG. **3** is a schematic illustration of an exemplary embodiment of the present invention, in particular provided for illustrating the general principle of the present invention. As shown in FIG. **3**, a stroke of connecting rod **50** of tamper device **40** may be adjusted by a rotational adjustment of a relative rotational positioning between an eccentric bushing **48** and an eccentric section **47** of the tamper shaft **46**, while the tamper shaft **46** is running, the rotational adjustment being performed in the way as described in detail with reference to FIGS. **2c** to **2f** above.

For providing the rotational adjustment of a relative rotational positioning between bushing **48** and section **47** of shaft **46**, the tamper device **40** of this embodiment comprises an inner-toothed hollow wheel gear train **60**. The gear train **60** is connected to the tamper shaft **46**, to the eccentric bushing **48** and to a drive force receiving element **70**. The drive force receiving element **70** is configured to be able to receive a drive force for driving the inner-toothed hollow wheel gear train **60** when the tamper shaft **46** is rotating. In the embodiment of FIG. **3**, the inner-toothed hollow wheel gear train **60** is realized as a planetary gear **60a**.

In the present application the term “inner-toothed hollow wheel gear train” is used for a gear train which comprises at least one inner-toothed hollow wheel. In the present application the term “inner-toothed hollow wheel” is used for a hollow wheel which comprises teeth on its inner surface, FIG. **3** shows examples of such inner-toothed hollow wheels in the form of a shaft side ring gear **62**, and a bushing side ring gear **64**, both described in detail below. The use of a planetary gear **60a** as an inner-toothed hollow wheel gear train **60** is an advantage since it is reliable and cost-efficient implementation of an inner-toothed hollow wheel gear train **60**. However, a person of ordinary skill in the art will appreciate that other inner-toothed hollow wheel gear trains **60** may be used.

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An advantage of the inventive tamper device **40** is that the stroke of the connecting rod **50** driving tamper bar **42** can be adjusted individually and arbitrarily. This is a strong advantage since an individual local position of the screed **18** might require an individual adjustment of the stroke **S**. If for example a road paver **10** uses more than one screed section, for example a basic screed and optionally screed extensions that can be extended at the basic screed for changing the working width of the road paver **10**, each of these components of the screed **18** may comprise its own tamper device **40**, so that the present invention enables an individual adjustment of the stroke **S** for each screed section. Due to the inventive use of an inner-toothed hollow wheel gear train **60** in the inventive tamper device **40** an adjustment of the stroke **S** can be provided during rotation of the tamper shaft **46**. It is therefore in particular not necessary to stop the tamper device **40** to adjust the stroke **S**. This is a strong advantage since therefore a costly stop of the work of the working paver **10** can be avoided.

The depicted planetary gear **60a** is a two-stage planetary gear **60a** and comprises a shaft side ring gear **62** being connected to the tamper shaft **46** in a torque-proof manner. This connection is realized by a disk **62a** being fitted into a corresponding circumferential recess **46b** in shaft **46** in a torque-proof manner, and the shaft side ring gear **62** being attached to disk **62a** by screws **6b**. Planetary gear **60a** further comprises a bushing side ring gear **64** being connected to the eccentric bushing **48** in a torque-proof manner. This connection may be realized by a having a shell-like structured housing **76** being integral with the eccentric bushing **48**. Not shown, housing **76** may be connected to the eccentric bushing **48** in a torque-proof manner for example by also using a screw connection. The bushing side ring gear **64** is attached to the housing **76** by screws **64b**.

Planetary gear **60a** further comprises a common planet gear **66** connecting the shaft side ring gear **62** with the bushing side ring gear **64**. This connection is realized by meshing with both of gears **62** and **64** in a way known to a person of ordinary skill in the art. Planetary gear **60a** further comprises a common sun gear **68** being connected to the drive force receiving element **70** in a torque-proof manner. This connection may be realized by a having a tube **74** surrounding shaft **46** and being integral with the disk-shaped drive force receiving element **70** and the central sun gear **68**.

Therefore, while housing **76** serves as a shell for the shaft side ring gear **62**, the bushing side ring gear **64**, the planet gear **66**, and the sun gear **68**, only the shaft side ring gear **62**, the planet gear **66**, and the sun gear **68** are rotatable relative to the housing **76**. The eccentric bushing **48** and the bushing side ring gear **64** are fixed to or integral with the housing **76** in a torque-proof manner.

By providing the housing **76** for the gears **62**, **64**, **66**, **68** and by simultaneously connecting the housing **76** to the eccentric bushing **48** of the tamper shaft **46** in a torque-proof manner it is provided a simple and reliable protection of the gears **62**, **64**, **66**, **68** in a possibly rough environment when the inventive tamper device **40** is for example used on a road paver **10**, while simultaneously the housing **76** is simply also part of the second stage or the output side of such planetary gear **60a**.

For further details of the structure of planetary gear **60a** and of the interaction of gears **62**, **64**, **66** and **68** of planetary gear **60a** it is also referred to FIG. **3a** showing a cross section through gears **62**, **66** and **68** along plane B-B in FIG. **3** and using dashed lines to show gear **64** not in the sectional plane B-B.

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As can be seen in FIG. **3a**, the shaft side ring gear **62** comprises 89 teeth **62t**, the bushing side ring gear **64** shown in dashed lines comprises 87 teeth **64t**, the planet gear **66** comprises 13 teeth **66t**, and the sun gear **68** comprises 62 teeth **68t**. Such ratio has been found as being particularly useful for the requirements of road pavers **10**. In particular, the shaft side ring gear **62** comprises more teeth **62t** than the bushing side ring gear **64**.

According to an alternative, not shown embodiment a ratio of a number of teeth **62t** of the shaft side ring gear **62** to a number of teeth **64t** of the bushing side ring gear **64** is about 30:1 to about 250:1. According to a further alternative, not shown embodiment the shaft side ring gear **62** comprises less teeth **62t** than the bushing side ring gear **64**. According to a further alternative, not shown embodiment a ratio of a number of teeth **62t** of the shaft side ring gear **62** to a number of teeth **64t** of the bushing side ring gear **64** should be about 1:30 to about 1:250. Such ranges of ratios have been found as being particularly useful for the requirements of road pavers **10**.

As shown in FIG. **3**, the drive force receiving element **70** is configured as a disk-like brake force receiving element being configured to be able to receive a brake force for driving the planetary gear **60a** when the tamper shaft **46** is rotating. Such brake force may be applied by a mechanical brake **70a** shown in simplified form and comprising brake shoes **70b** which can be moved laterally against the disk-like drive force receiving element **70**. For that purpose the brake force receiving element comprises a disk-like adjustment wheel **72** sitting on the tamper shaft **46**. The adjustment wheel **72** is rotatable relative to the tamper shaft **46**. Since the adjustment wheel **72** is connected to the sun gear **68** in a torque-proof manner via connecting tube **74**, the adjustment wheel **72** is configured and enabled to drive the eccentric bushing **48** in a direction against a direction of rotation of the tamper shaft **46**, in case the adjustment wheel **72** is receiving a brake force when the tamper shaft **46** is rotating.

An adjustment of the stroke of the connecting rod **50** while the tamper shaft **46** is rotating, works as follows: Providing a certain brake force to the adjustment wheel **72** causes a corresponding reduction of the rotational speed of the adjustment wheel **72**. Since the adjustment wheel **72** can rotate relative to the tamper shaft **46**, this also causes a corresponding reduction of the rotational speed of the adjustment wheel **72** relative to an unchanged rotational speed of the tamper shaft **46**. Since sun gear **68** is integral with adjustment wheel **72** via tube **74**, this also causes a corresponding reduction of the rotational speed of the sun gear **68** relative to the rotational speed of the tamper shaft **46**.

Since the tamper shaft **46** is fixed to the shaft side ring gear **62**, this causes also a corresponding reduction of the rotational speed of the sun gear **68** relative to the rotational speed of the shaft side ring gear **62**.

Since the sun gear **68** is connected to the shaft side ring gear **62** via the planet gear **66**, this causes the planet gear **66** to move between sun gear **68** and shaft side ring gear **62**. Since the planet gear **66** extending into the bushing side ring gear **64**, also, and since the bushing side ring gear **64** has less teeth **64t** than the shaft side ring gear **62**, the planet gear **66** forces the bushing side ring gear **64** into a rotation relative to the shaft side ring gear **62**.

Since the bushing side ring gear **64** is fixed to eccentric bushing **48**, this causes a corresponding rotation of eccentric

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bushing 48. As discussed above, a rotation of eccentric bushing 48 causes a corresponding adjustment of the stroke S of connecting rod 50.

Preferably, a shaft eccentricity of the eccentric section 47 and a bushing eccentricity of the eccentric bushing 48 are configured so that the stroke is adjustable between a predefined minimum, e.g. zero, and a predefined maximum. Preferably, if the predefined minimum is zero, this embodiment provides the possibility to not only arbitrarily adjust the stroke S between a predefined non-zero minimum and a predefined maximum but to adjust the stroke S even to zero. Such a possibility may be useful in particular if a screed 18 uses two parallel tamper bars 42, 44, for example one tamper bar 42 being mounted at the ends 52 of two connecting rods 50 having an adjustable stroke S according to the present invention, and another tamper bar 44 having a fixed stroke. By a minimum stroke being zero it would then be possible to adjust the stroke of the upstream tamper bar 42 to zero, if for example the impact of the stroke S of the second downstream tamper bar 44 would be sufficient for a certain working condition of a respective road paver 10.

By providing a brake force receiving element 70 receiving a brake force for driving the gear 60a it is not only possible to provide the rotational adjustment of a relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 while the tamper shaft 46 is rotating or running, but it is additionally possible to use the rotational energy of the rotating tamper shaft 46 for driving the inventive gear train 60a. By for example connecting the brake force receiving element 70 with the input side of the inventive gear train 60a as described above the brake force receiving element 70 may simply rotate together with the whole gear train 60a which itself is rotating together with the tamper shaft 46 due to its connection to the tamper shaft 46. Only if an adjustment of the relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 for an adjustment of the stroke S is desired, a brake force is provided to the brake force receiving element 70 connected with sun gear 68 on the input side of the inventive gear train 60a to drive the inventive gear train 60a and to thereby provide the rotational adjustment of a relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 of running shaft 46. By using an adjustment wheel 72 rotatable sitting on tamper shaft 46, a simple and cost-efficient implementation of a brake force receiving element 70 is provided. By connecting the adjustment wheel 72 with the sun gear 68 of a planetary gear 60a as described above a simple and cost-efficient implementation of a connection between the adjustment wheel 72 and a planetary gear 60a is provided since the sun gear 68 is the inner most part of a planetary gear 60a so that it is in particular simple to for example use a connecting tube 74 as discussed above to connect adjustment wheel 72 and sun gear 68.

FIG. 4a is an illustration of another embodiment of the present invention. As shown in FIG. 4a, tamper device 40 comprises a second set of non-adjustable connecting rods 101, 102 which may be connected with the downstream tamper bar 44 of the screed 18 of FIG. 1 as described above. FIG. 4a also comprises a further brake force receiving element 80 being connected with the brake force receiving element 70 is configured to be able to receive a brake force for driving the gear unit 81 by comprising a further adjustment wheel 84 sitting on the tamper shaft 46. The further adjustment wheel 84 is rotatable relative to the tamper shaft 46, whereby the further adjustment wheel 84 is configured

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to drive the adjustment wheel 72 in a direction of rotation of the tamper shaft 46, in case the further adjustment wheel 84 is receiving a brake force when the tamper shaft 46 is rotating.

As shown in FIG. 4a the gear unit 81 comprises a friction gear 83. The friction gear 83 comprises at least one ball 83a which connects adjustment wheel 72 with further adjustment wheel 80 in a force-fit manner. The at least one ball 83a is held rotatable in a central gear element 83b which is fixedly mounted on tamper shaft 46 or tube 88, see FIG. 5 showing a cross-sectional view of FIG. 4b. Additionally, friction gear 83 comprises a biasing element 83c which can be axially moved along tamper shaft 46 and which can be fixed in a desired biasing position on tamper shaft 46 to provide a biasing force via a spring 83d to the further adjustment wheel 84.

The embodiment of FIGS. 4a and 5 works as follows: In case a brake force is applied to the further adjustment wheel 84, for example by a mechanical brake, or for example induced in the further adjustment wheel 84 by an eddy current brake 86 as shown in FIG. 4b, the further adjustment wheel 84 will be slowed down in its rotational velocity in relation to the rotational velocity of the tamper shaft 46. This causes a rotation of the ball 83a being in friction-fit contact with the further adjustment wheel 84. Since for example a rotation of the ball 83a into the paper plane of FIG. 4a on the right hand side of gear element 83b means a rotation of ball 83a out of the paper plane of FIG. 4a on the left hand side of gear element 83b, the adjustment wheel 72 is forced by ball 83a to rotate in the same direction as the tamper shaft 46. This causes a respective rotation of the eccentric bushing 48 and therefore results in an adjustment of the stroke S of connecting rod 50 and a corresponding tamper bar as discussed above.

In case a friction force by eddy current brake 86 is induced in the adjustment wheel 72, ball 83a will also translate this in an anti-rotation of the further adjustment wheel 84. This however has no consequence for the stroke of the tamper device 40 since such a rotation only causes a rotation of the further adjustment wheel 84 which is not connected to any bushing.

According to another embodiment shown in FIG. 4b the gear unit 81 comprises a cog wheel 82 replacing the friction gear 83 of FIG. 4a. The cog wheel 82 is sitting on a laterally extending projection and can rotate in the paper plane of FIG. 4b. The cog wheel 82 is meshing with its teeth 82t in corresponding openings in both adjustment wheels 72 and 84.

The embodiment of FIG. 4b works as follows: In case a brake force is applied to the further adjustment wheel 84, the further adjustment wheel 84 will be slowed down in its rotational velocity in relation to the rotational velocity of the tamper shaft 46. This causes a rotation of the cog wheel 82. This causes the adjustment wheel 72 to rotate. This causes a respective rotation of the eccentric bushing 48 as discussed above and therefore results in an adjustment of the stroke S of connecting rod 50 and a corresponding tamper bar as discussed above.

By providing a further brake force receiving element, e.g. in form of a further adjustment wheel 84, in addition to the brake force receiving element 70, and by connecting the further brake force receiving element with the brake force receiving element 70 via a gear unit 81, it is advantageously possible to provide a brake force to the further adjustment wheel 84 which brake force is then translated by the gear unit 81 so that the adjustment wheel 72 can be rotated in the same direction as the rotation direction of the tamper shaft

46. Accordingly, it is possible by using two adjustment wheels 72, 84 connected via gear unit 81 to either brake on the adjustment wheel 72 to drive the eccentric bushing 48 in a direction against a direction of rotation of the tamper shaft 46, or to brake on the further adjustment wheel 84 so that its rotation is slowed down with respect to the rotation of the tamper shaft 46, this slowing down of the further adjustment wheel 84 being translated by the gear unit 81 into an acceleration of the adjustment wheel 72 which drives the eccentric bushing 48 in a direction of rotation of the tamper shaft 46. With other words: By the present embodiment the direction of rotational adjustment of a relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 can be chosen by selecting either the adjustment wheel 72 or the further adjustment wheel 84 to be braked on. Thereby a necessity to provide a rotational adjustment of a relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 of more than 180° is advantageously avoided. Accordingly, an adjustment on the basis of this embodiment substantially reduces the average time for rotational adjustment.

According to the embodiment of FIGS. 4b and 5 the tamper shaft 46 comprises a further eccentric section 49 with a further eccentric bushing 51 mounted on the further eccentric section 49, and a further connecting rod 53 being rotatable mounted on the further eccentric bushing 51 for being driveable with stroke motions having a stroke. The stroke is adjustable by an rotational adjustment of a relative rotational positioning between the further eccentric bushing 51 and the further eccentric section 49 in the same fashion as discussed above with respect to connecting rod 50.

By this embodiment it is possible to in parallel adjust the stroke S of the connecting rod 50 and the further connecting rod 53 by connecting the eccentric bushings 48, 51 of each of the connecting rods 50, 53 so that a relative rotational positioning of the eccentric bushing 48 according to the present invention causes a parallel additional positioning of the further eccentric bushing 51.

As shown in FIG. 5, the eccentric bushing 48 and the further eccentric bushing 51 are connected in a torque-proof manner by an elongated tube 88 surrounding the tamper shaft 46 between the eccentric bushing 48 and the further eccentric bushing 51.

As can be seen in FIG. 5, the elongated tube 88 is mounted at each bushing 48, 51 with lateral play to compensate a lateral movement of the bushings 48, 51 due to the eccentricity of the respective eccentric sections 47, 49 on the tamper shaft 46 when the stroke being adjusted by an rotational adjustment of a relative rotational positioning between the eccentric bushings 48, 51 and the eccentric sections 47, 48.

By mounting the elongated tube 88 connecting the eccentric bushing 48 with the further eccentric bushing 51 at each bushing 48, 51 with lateral play, this embodiment advantageously compensates a lateral movement of the bushings 48, 51 due to the eccentricity of the respective eccentric sections 47, 49 on the tamper shaft 46 when the stroke S being adjusted by an rotational adjustment of a relative rotational positioning between the eccentric bushings 48, 51 and the eccentric sections 47, 49 of the tamper shaft 46.

FIG. 6 is a perspective view of FIG. 5. FIG. 7 corresponds to FIG. 6 but does only show a partial cross-section. FIG. 8 corresponds to FIG. 7 but the part showing the cross-section has even further being reduced.

FIG. 9 corresponds to FIG. 8 but shows only a part of FIG. 8 in that a cut through the housing 76 of the gear train 60 has been performed and everything left of the cut has been

omitted to have a view on elements of the gear train 60 in the housing 76. As also indicated in FIG. 5 a hollow ring 63 is fixedly mounted in a torque-proof manner to tamper shaft 46. Additionally, the hollow ring 63 is fixedly connected by screws 65 with the shaft side ring gear 62 so that the shaft side ring gear 62 is therefore also fixedly connected in a torque-proof manner with tamper shaft 46. As in particular can be seen in FIG. 9, the hollow ring 63 carries a projection 67 radially pointing to the tamper shaft 46 and being in contact with the surface of tamper shaft 46 and also been fixedly connected by a radially mounted screw 69 to the tamper shaft 46 in a torque-proof manner.

As can be seen in respective side views of FIG. 9, depicted in FIGS. 10a and 10b, elongated connecting tube 88 carries a projection 88a extending in axial direction of tamper shaft 46 into an opening 91 where the projection 67 of hollow ring 63 does not have contact with tamper shaft 46 but radially is spaced from the surface 46a of tamper shaft 46. As can be seen in FIG. 9 the projection 88a has a circumferential extension of about half the circumference of the tube 88. Due to the projection 88a into the opening 91 between hollow ring 63 and tamper shaft 46 at the axial position of projection 67, tube 88 can only be rotated between the tube positions shown in FIGS. 10a and 10b.

An alternative, not shown embodiment comprises an electric motor, and the drive force receiving element 70 of the afore-mentioned embodiments is configured to be driven by the electric motor. By using an electric motor, the drive force receiving element 70 can receive a drive force for driving the inner-toothed hollow wheel gear train 60 when the tamper shaft 46 is rotating, as well as when the tamper shaft 46 is standing still.

According to an alternative, not shown embodiment the inner-toothed hollow wheel gear train 60 of the afore-mentioned embodiments comprises a strain wave gear. Preferably, such strain wave gear comprises a circular spline being connected to the tamper shaft 46 in a torque-proof manner, a flex spline being connected to the eccentric bushing 48 in a torque-proof manner, and a drive element for moving the flex spline and being connected to the drive force receiving element 70 in a torque-proof manner.

In case the drive force receiving element 70 of this embodiment comprises a brake force receiving element being configured to be able to receive a brake force for driving the strain wave gear when the tamper shaft 46 is rotating, the brake force receiving element may be configured to be able to receive a brake force by comprising an adjustment wheel 72 sitting on the tamper shaft 46, the adjustment wheel 72 being rotatable relative to the tamper shaft 46, and the adjustment wheel 72 being connected to the flex spline in a torque-proof manner, preferably via a connecting tube 74 surrounding the tamper shaft 46, whereby the adjustment wheel 72 is being configured to drive the eccentric bushing 48 in a direction against a direction of rotation of the tamper shaft 46, in case the adjustment wheel 72 is receiving a brake force when the tamper shaft 46 is rotating.

The use of a strain wave gear as gear 60, also known as a harmonic gear or harmonic drive, provides the advantages of nearly no backlash, enhanced compactness, reduced weight, high gear ratios, reconfigurable ratios within the same housing, enhance resolution and excellent repeatability when repositioning initial loads, and a high torque capability. In particular, the use of a strain wave gear provides high gear reduction ratios in a small volume. As an example, it is possible to realize a gear reduction ratio from 30:1 up to

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320:1 in the same space in which a normal planetary gear 60a typically produces a reduction ratio of 10:1.

According to a second aspect of the present invention, FIG. 11 shows a schematic illustration of an embodiment of a method for adjusting a stroke of a tamper device 46 for a screed 18 of a road paver 10. The tamper device 40 comprises a rotatable driveable tamper shaft 46 comprising an eccentric section 47, an eccentric bushing 48 mounted on the eccentric section 47, and a connecting rod 50 being rotatable mounted on the eccentric bushing 48 for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing 48 and the eccentric section 47. The method comprises the step 300 of providing the rotational adjustment by driving an inner-toothed hollow wheel gear train 60 by the step 200 of providing a drive force to a drive force receiving element 70. The drive force receiving element 70 is connected to the inner-toothed hollow wheel gear train 60. The drive force receiving element 70 is configured to be able to receive the drive force for driving the inner-toothed hollow wheel gear train 60 when the tamper shaft 46 is rotating. The inner-toothed hollow wheel gear train 60 is also connected to the tamper shaft 46 and to the eccentric bushing 48.

Preferably, when executing the method, the method can comprise the further step of measuring an angle of the relative rotational positioning between the eccentric bushing 48 and the eccentric section 47. Hereby an exact information about the relative rotational positioning between the eccentric bushing and the eccentric section can be retrieved.

Preferably, when executing the method, the method can comprise the further step of using the measured angle of the relative rotational positioning to define the amount of rotational adjustment of the relative rotational positioning between the eccentric bushing 48 and the eccentric section 47. Hereby a precise adjustment of the relative rotational positioning between the eccentric bushing 48 and the eccentric section 47 and thereby a precise determination of the stroke S can be achieved. This possibility is particularly advantageous if such method is for example used in combination with a measurement of a laid material thickness of the material paved by paver 10, and/or of a measurement of compaction of the material paved by paver 10, so that the stroke S can immediately be adjusted on the basis of the measured parameters of material thickness and/or compaction and the actual stroke derived from the actually measured angle of the relative rotational positioning as mentioned above.

Preferably, when executing the method, at least one computer program may be used, the at least one computer program comprising program code means for performing the steps of the method for adjusting a stroke of a tamper device 40 for a screed 18 of a working machine as described herein, when said program is run on at least one computer.

A computer readable medium can be provided, the computer readable medium carrying at least one computer program comprising program code means for performing the steps of the method for adjusting a stroke of a tamper device 40 for a screed 18 of a working machine as described herein, when said program product is run on at least one computer.

As shown in FIG. 1, a control unit 100 can be provided for controlling a tamper device 40 for a screed 18 of a working machine, in particular a paver 10, the control unit 100 being configured to perform the steps of the method for adjusting a stroke of a tamper device 40 for a screed 18 of a working machine as described herein.

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Preferably, the control unit 100 comprises a mechanical and/or electrical sensor for measuring an angle of the relative rotational positioning between the eccentric bushing 48 and the eccentric section 47.

The present invention also relates to a working machine, in particular a road paver 10, comprising a control unit 100 as described herein.

It is to be understood that the present invention is not limited to the embodiments described above and illustrated in the drawings; rather, the skilled person will recognize that many changes and modifications may be made within the scope of the appended claims.

The invention claimed is:

1. A tamper device for a screed of a working machine comprising a paver, the tamper device comprising:

a rotatable driveable tamper shaft comprising an eccentric section,

an eccentric bushing mounted on the eccentric section,

a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke,

the stroke (S) being adjustable by a rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section,

an inner-toothed hollow wheel gear train for providing the rotational adjustment,

the gear train being connected to the tamper shaft, to the eccentric bushing, and to a drive force receiving element configured to receive a drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating.

2. The tamper device of claim 1, wherein the inner-toothed hollow wheel gear train comprises a planetary gear.

3. The tamper device of claim 2, wherein the planetary gear is a two-stage planetary gear and comprises

a shaft side ring gear being connected to the tamper shaft in a torque-proof manner,

a bushing side ring gear being connected to the eccentric bushing in a torque-proof manner,

a common planet gear connecting the shaft side ring gear with the bushing side ring gear, and

a common sun gear being connected to the drive force receiving element in a torque-proof manner.

4. The tamper device of claim 3, further comprising a housing for the shaft side ring gear, the bushing side ring gear, the planet gear, and the sun gear,

the housing being connected to the eccentric bushing in a torque-proof manner, the bushing side ring gear being connected to the housing in a torque-proof manner, whereas the shaft side ring gear, the planet gear, and the sun gear being rotatable relative to the housing.

5. The tamper device of claim 3, wherein

the shaft side ring gear comprises less teeth than the bushing side ring gear, preferably a ratio of a number of teeth of the shaft side ring gear to a number of teeth of the bushing side ring gear being about 1:30 to about 1:250; or

the shaft side ring gear comprises more teeth than the bushing side ring gear, preferably a ratio of a number of teeth of the shaft side ring gear to a number of teeth of the bushing side ring gear being about 30:1 to about 250:1;

wherein the shaft side ring gear comprises 87 teeth and the bushing side ring gear (64) comprises 89 teeth, the planet gear comprises 13 teeth, and the sun gear comprises 62 teeth.

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6. The tamper device of claim 1, wherein the drive force receiving element comprises a brake force receiving element being configured to be able to receive a brake force for driving the inner toothed hollow wheel gear train when the tamper shaft is rotating.

7. The tamper device of claim 6, wherein the brake force receiving element is configured to be able to receive a brake force by an adjustment wheel sitting on the tamper shaft, the adjustment wheel being rotatable relative to the tamper shaft, and

the adjustment wheel connected to the inner toothed hollow wheel gear train in a torque-proof manner, preferably the adjustment wheel being connected to the sun gear in a torque-proof manner via a connecting tube surrounding the tamper shaft,

whereby the adjustment wheel is configured to drive the eccentric bushing in a direction against a direction of rotation of the tamper shaft when the adjustment wheel is receiving a brake force when the tamper shaft is rotating.

8. The tamper device of claim 6, comprising a further brake force receiving element connected with the brake force receiving element via a gear unit,

the further brake force receiving element being configured to receive a brake force for driving the gear unit by a further adjustment wheel sitting on the tamper shaft, the further adjustment wheel being rotatable relative to the tamper shaft,

whereby the further adjustment wheel is configured to drive the adjustment wheel in a direction of rotation of the tamper shaft when the further adjustment wheel is receiving a brake force when the tamper shaft is rotating.

9. The tamper device of claim 8, the gear unit comprising at least one of a cog wheel and a friction gear.

10. The tamper device of claim 6, comprising at least one of a mechanical brake and an eddy current brake for providing a brake force to the brake force receiving element and/or the further brake force receiving element.

11. The tamper device of claim 1, further comprising a tamper bar mounted at an end of the connecting rod.

12. The device of claim 1, wherein a shaft eccentricity of the eccentric section and a bushing eccentricity of the eccentric bushing are configured so that the stroke is adjustable between a predefined minimum, and a predefined maximum.

13. The tamper device of claim 1, wherein the tamper shaft comprises

a further eccentric section with a further eccentric bushing mounted on the further eccentric section, and

a further connecting rod being rotatable mounted on the further eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by a rotational adjustment of a relative rotational positioning between the further eccentric bushing and the further eccentric section,

wherein the eccentric bushing and the further eccentric bushing are connected in a torque-proof manner by an elongated tube surrounding the tamper shaft between the eccentric bushing and the further eccentric bushing, the elongated tube mounted at each bushing with lateral play to compensate a lateral movement of the bushings due to the eccentricity of the respective eccentric sections on the tamper shaft, when the stroke being adjusted by a rotational adjustment of a relative rotational positioning between the eccentric bushings and the eccentric sections.

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14. The tamper device of claim 1, comprising an electric motor, and wherein the drive force receiving element is configured to be driven by the electric motor.

15. The tamper device of claim 1, wherein the inner-toothed hollow wheel gear train comprises a strain wave gear.

16. The tamper device of claim 15, wherein the strain wave gear comprises

a circular spline being connected to the tamper shaft in a torque-proof manner,

a flex spline connected to the eccentric bushing in a torque-proof manner, and

a drive element for moving the flex spline and connected to the drive force receiving element in a torque-proof manner.

17. The tamper device of claim 16, wherein the drive force receiving element comprises a brake force receiving element being configured to receive a brake force for driving the strain wave gear when the tamper shaft is rotating,

the brake force receiving element configured to receive a brake force by comprising an adjustment wheel sitting on the tamper shaft,

the adjustment wheel rotatable relative to the tamper shaft and being connected to the flex spline in a torque-proof manner via a connecting tube surrounding the tamper shaft,

whereby the adjustment wheel is configured to drive the eccentric bushing in a direction against a direction of rotation of the tamper shaft when the adjustment wheel is receiving a brake force when the tamper shaft is rotating.

18. A screed of a working machine comprising a paver, comprising a tamper device according to claim 1.

19. A working machine comprising a road paver, comprising a screed according to claim 18.

20. A method for adjusting a stroke of a tamper device according to claim 1, for a screed of a working machine comprising a paver, the tamper device comprising: a rotatable driveable tamper shaft comprising an eccentric section, an eccentric bushing mounted on the eccentric section, a connecting rod being rotatable mounted on the eccentric bushing for being driveable with stroke motions having a stroke, the stroke being adjustable by an rotational adjustment of a relative rotational positioning between the eccentric bushing and the eccentric section, the method comprising:

providing the rotational adjustment by driving an inner-toothed hollow wheel gear train by providing a drive force to a drive force receiving element connected to the inner-toothed hollow wheel gear train and being configured to be able to receive the drive force for driving the inner-toothed hollow wheel gear train when the tamper shaft is rotating,

the inner-toothed hollow wheel gear train also being connected to the tamper shaft and to the eccentric bushing.

21. A method according to claim 20, further comprising measuring an angle of the relative rotational positioning between the eccentric bushing and the eccentric section.

22. A method according to claim 21, further comprising using the measured angle of the relative rotational positioning to define the amount of rotational adjustment of the relative rotational positioning between the eccentric bushing and the eccentric section.

23. A computer program comprising program code means for performing the steps of claim 20 when said program is run on a computer.

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24. A computer readable medium carrying a computer program comprising program code means for performing the steps of claim **20** when said program product is run on a computer.

25. A control unit for controlling a tamper device for a 5
screed of a working machine comprising a paver, the control unit being configured to perform the steps of the method according to claim **20**.

26. The control unit of claim **25**, further comprising a mechanical and/or electrical sensor for measuring an angle 10
of the relative rotational positioning between the eccentric bushing and the eccentric section.

27. A working machine comprising a road paver, comprising a control unit according to claim **25**.

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